

Project ARIA: A Cognitive Multi-Agent Vision-Language-Action Architecture for Autonomous Manipulation on Low-Cost 5-DoF Robotic Arms

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Abstract—Autonomous robotic manipulation has entered a transformative era driven by Vision-Language-Action (VLA) foundation models. However, contemporary robotic foundation policies are monolithic black boxes demanding datacenter-scale compute (> 24 GB VRAM), incurring prohibitive inference latencies (> 100 ms), and failing catastrophically on low-cost underactuated manipulators. In this paper, we present the research formulation, architectural design, physical implementation, and empirical validation of Project ARIA (Autonomous Reasoning & Interaction Agent), a decentralized cognitive multi-agent VLA architecture specifically engineered to deliver high-precision physical autonomy on budget 5-DoF manipulators within an 8.0 GB VRAM budget. ARIA bridges the gap between semantic cognition and deterministic motor execution through a stratified hierarchy of fifteen lifecycle-managed software agents communicating asynchronously across a unified State Bus. To eliminate costly active depth cameras, we formulate a Zero-Cost-Depth perception pipeline combining overhead YOLOv8m detection and SAM2 segmentation with wrist-mounted Depth-Anything v2 grounded via physical claw geometry ($RMSE = 8.2$ mm). For deterministic execution, we derive an exact closed-form analytical Inverse Kinematics solver resolving the 5-DoF anthropomorphic chain in 0.08 ms ($231\times$ faster than numerical optimization) with 100% reachability determinism. Cognitive reasoning is orchestrated by an open-weight quantized LLM executing Tree-of-Thoughts task decomposition with an emoji-annotated Chain-of-Thought telemetry stream and Human-in-the-Loop confirmation gates. Benchmarked across 10 standardized manipulation tasks, ARIA achieves an 89.0% overall success rate, substantially outperforming state-of-the-art foundation policies including LeRobot ACT (72.0%), OpenVLA (68.0%), and Physical Intelligence π_0 (61.0%). In automated conveyor workcells, ARIA achieves a 93.3% dynamic sorting rate with feedforward velocity matching ($\Delta v \leq 0.002$ m/s) and sub-millimeter palletizing precision ($RMSE = 1.14$ mm), slashing system deployment costs by over 99%.

Index Terms—Cognitive robotics, multi-agent systems, vision-language-action (VLA) models, low-cost manipulators, zero-cost depth estimation, inverse kinematics, conveyor tracking, explainable artificial intelligence.

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Supplementary downloadable multimedia material is submitted with this paper, demonstrating multi-agent execution, Gazebo digital twin simulation, dynamic conveyor sorting, and closed-form IK control.

I. Introduction

THE realization of autonomous physical agents capable of executing complex manipulation in unstructured human environments represents a foundational ambition of robotics and embodied artificial intelligence [1]. Recent advances in dynamic trajectory planning, visual servoing, and predictive object interception established by Sun et al. [1] in IEEE Access within ROS and Gazebo simulation frameworks have demonstrated that accurate kinematic forecasting is essential for grasping moving targets. However, traditional visual servoing frameworks rely on static, hand-engineered geometric primitives that fail to generalize across diverse operational contexts, highlighting the critical imperative for architectures capable of adaptive semantic reasoning and autonomous recovery.

To endow manipulators with generalizable semantic comprehension, the robotics community has increasingly embraced large-scale Vision-Language-Action (VLA) foundation models. As comprehensively surveyed by the Embodied AI Review Group [2] in Robotics and Autonomous Systems, contemporary VLA policies leverage massive cross-embodiment datasets to enable zero-shot semantic generalization across novel visual objects. Nevertheless, existing foundation policies operate as monolithic autoregressive transformers requiring over 16 GB of dedicated VRAM and incurring control loop inference latencies exceeding 140 ms per forward pass. On physical robotic hardware, this latency induces severe closed-loop instability, preventing responsive reaction to dynamic environmental events or sudden mechanical slippage.

To establish foundational sensorimotor representations that transcend monolithic end-to-end black boxes, Kroeimer et al. [3] in the Annual Review of Control, Robotics, and Autonomous Systems formalized the algorithmic structures governing modern robotic manipulation. Their analysis reveals that robust physical interaction mandates a principled stratification between high-level cognitive task planning, mid-level geometric state estimation, and high-frequency deterministic motor control. Monolithic end-to-end foundation models conflate these distinct physical hierarchies, leading to catastrophic action divergence when minor out-of-distribution visual perturbations occur.

This structural disconnect is exacerbated during im-

itation learning and policy acquisition. Ravichandar et al. [4] in the Annual Review of Control, Robotics, and Autonomous Systems reviewed the core bottlenecks of Robot Learning from Demonstration (LfD), highlighting that imitation learning frameworks assume expensive research-grade 7-DoF manipulators (e.g., Franka Emika Panda, Kinova Gen3, UR5e costing \$25,000–\$60,000) equipped with ideal joint-torque feedback and negligible backlash. Conversely, accessible 5-DoF educational and light-industrial manipulators (< \$150) driven by PWM hobbyist servomotors remain systematically neglected due to structural compliance, joint elasticity, deadbands, and absence of native torque sensing.

On the middleware and system orchestration level, distributed coordination is essential for multi-agent embodied autonomy. Macenski et al. [5] in Science Robotics established the architectural foundations of the Robot Operating System 2 (ROS 2), demonstrating how deterministic Data Distribution Service (DDS) Quality of Service (QoS) profiles, real-time publish-subscribe topics, and managed node lifecycles provide the determinism required for physical autonomy. However, contemporary embodied VLA models remain completely isolated from ROS 2 lifecycle management, running as unmonitored Python scripts that lack fault-recovery protocols and hard real-time safety gating.

To address the limitations of centralized coordination in complex manipulation arenas, Muthusamy et al. [6] in Frontiers in Robotics and AI introduced a multi-robot collaborative manipulation framework utilizing real-time distributed optimization. Their work establishes that distributed multi-agent topologies eliminate single-point computational bottlenecks and improve disturbance rejection during coordinated grasping. Translating this decentralized consensus paradigm to modular cognitive agents operating on a single resource-constrained edge computer presents an unexplored opportunity to deliver modular, transparent embodied intelligence.

A parallel challenge lies in the visual perception domain, where physical manipulation depends critically on accurate 3D scene geometry. In standard laboratory setups, active structured-light or Time-of-Flight (ToF) RGB-D sensors (e.g., Intel RealSense D435i, Photoneo) are universally deployed. However, as comprehensively documented by Jiang et al. [7] in the IEEE Transactions on Artificial Intelligence, active infrared depth sensors suffer from severe physical failure modes, including total signal absorption or multi-path specular scattering on transparent glassware, glossy metallic parts, and reflective containers. Crucially, active depth cameras exhibit an intrinsic minimum-range blind zone (< 0.20 m), leaving the manipulator entirely blind during the final, high-precision grasping approach.

To circumvent active depth sensor failures, monocular RGB depth estimation has emerged as a promising alternative. As demonstrated by Chen et al. [8] in Sensors, adaptive grasp pose synthesis from low-cost optical cameras can achieve robust grasping if spatial uncertainty

is explicitly modeled. Nonetheless, zero-shot monocular depth networks generate normalized relative disparity fields rather than physical metric distance. Without an absolute physical reference or known geometric anchors, raw monocular disparity cannot be translated into metric Cartesian trajectory setpoints, creating a fundamental scale ambiguity bottleneck.

On the kinematic control layer, general-purpose robotics frameworks (such as MoveIt2, KDL, and TRAC-IK) rely on numerical optimization routines utilizing damped least-squares (DLS) or iterative Jacobian pseudoinverse techniques. As rigorously analyzed by Wagaa et al. [9] in Engineering Applications of Artificial Intelligence, numerical Inverse Kinematics (IK) solvers exhibit severe algorithmic vulnerabilities on articulated serial chains lacking spherical wrists, incurring 10–20 ms calculation latencies, getting trapped in local minima, and diverging near kinematic singularities. On underactuated 5-DoF arms where the end-effector roll and pitch axes are geometrically coupled with arm position, rank deficiency in the 6×5 geometric Jacobian causes standard numerical solvers to fail on boundary targets, demanding exact closed-form algebraic formulations.

Finally, in dynamic manufacturing environments, manipulators must track, intercept, and sort items moving on continuous industrial conveyor belts. Sahu et al. [10] in Scientific Reports demonstrated autonomous visual object tracking on an articulated manipulator within Gazebo simulation, highlighting the critical role of predictive intercept calculation. However, translating dynamic conveyor interception to multi-DoF articulated arms while performing real-time quality defect inspection and feedforward velocity matching under edge compute constraints remains an open challenge.

A. Research Gaps Identified

Synthesizing the state of the art reveals five fundamental, unresolved research gaps:

- 1) Gap 1: The Compute-Edge Disconnect in VLA Deployments: Existing foundation models mandate datacenter-grade multi-GPU infrastructure (> 24–80 GB VRAM). No unified cognitive architecture exists that partitions foundation capabilities into modular cooperating agents executing deterministically on a single consumer laptop GPU (< 8.0 GB VRAM).
- 2) Gap 2: Black-Box Opacity vs. Industrial Causal Diagnostics: Monolithic policies map pixels directly to motor actions without exposing intermediate spatial reasoning, inverse kinematics health, or contact physics. Consequently, when manipulation fails, operators cannot diagnose the root cause or trigger automated recovery.
- 3) Gap 3: Close-Range Sensor Dead-Zones and Metric Scale Ambiguity: Active structured-light depth cameras fail below 0.2 m and scatter on specular surfaces, while state-of-the-art monocular depth models pro-

vide only unscaled relative disparity maps lacking metric scale for Cartesian trajectory generation.

- 4) Gap 4: Numerical IK Instability on Underactuated 5-DoF Chains: General-purpose numerical IK solvers diverge near kinematic singularities and incur 10–20 ms latencies on 5-DoF arms lacking independent wrist roll/yaw axes, creating a need for deterministic sub-millisecond analytical solutions.
- 5) Gap 5: Dynamic Conveyor Rendezvous Under Budget Constraints: Prior moving-target manipulation literature assumes high-torque industrial 6/7-DoF arms backed by optical PLC shaft encoders. Achieving dynamic moving conveyor tracking ($\mathbf{v}_{ee} \approx \mathbf{v}_{belt}$) and dual-bin sorting on budget servo-driven arms remains completely unexplored.

B. Contributions of This Work

To bridge these gaps, this article presents Project ARIA (Autonomous Reasoning & Interaction Agent), establishing the following core scientific and engineering contributions:

- 1) Decentralized 15-Agent Cognitive Architecture: We design, implement, and validate a non-monolithic multi-agent architecture where 15 specialized agents operate as ROS 2 LifecycleNode instances across six functional tiers, guaranteeing fault isolation, modularity, and sub-5 ms emergency preemption within an 8.0 GB VRAM budget.
- 2) Zero-Cost-Depth Perception Pipeline: We formulate a markerless perception pipeline fusing overhead eye-to-hand RGB video (YOLOv8m detection, SAM2 instance segmentation) with wrist-mounted eye-in-hand Depth-Anything v2 monocular depth, mathematically grounded via physical claw geometry ($Z_{tips} = 0.065$ m) to achieve sub-centimeter metric accuracy ($RMSE = 8.2$ mm) without active infrared depth sensors.
- 3) Sub-Millisecond Closed-Form Analytical Kinematics: We derive an exact algebraic closed-form Inverse Kinematics solver resolving the 2R planar subproblem for 5-DoF anthropomorphic arms in 0.08 ms ($231\times$ faster than numerical optimization) with 100% reachability determinism and zero singularity divergence.
- 4) Hierarchical LLM Planning with Explainable CoT Telemetry: We integrate local quantized LLMs (Mistral-7B / Llama 3.1 8B via Ollama) utilizing Tree-of-Thoughts task decomposition with dynamic confidence scoring (τ), streaming an emoji-annotated Chain-of-Thought (CoT) telemetry log and triggering Human-in-the-Loop (HITL) safety confirmation whenever plan ambiguity arises.
- 5) Dynamic Conveyor Interception & Dual-Bin Sorting: We synthesize predictive rendezvous trajectories ($\mathbf{p}_{int}(t) = \mathbf{p}_0 + \mathbf{v}_{belt}t$) with feedforward velocity matching ($\mathbf{v}_{ee} \approx \mathbf{v}_{belt} = 0.05$ m/s, $\Delta v \leq 0.002$ m/s) to achieve zero-slip dynamic grasping and automated

sorting into a 4-pocket palletizing tray ($RMSE = 1.14$ mm) and scrap reject bin within a 3.8 s cycle.

- 6) Comprehensive Sim-to-Real Benchmark Validation: We benchmark ARIA across 10 standardized manipulation tasks, achieving an 89.0% overall success rate outperforming LeRobot ACT (72.0%), OpenVLA (68.0%), and π_0 (61.0%) while deploying an open-source glassmorphism web dashboard for real-time control.

The remainder of this article is organized as follows: Section II provides a detailed literature review across ten core robotics themes with a comprehensive summary matrix. Section III formalizes the physical manipulator platform and analytical kinematic derivations. Section IV expounds upon the decentralized multi-agent system architecture and State Bus. Section V details the hybrid Zero-Cost-Depth perception pipeline and grasp synthesis. Section VI presents cognitive task planning, reasoning, and explainable dialogue. Section VII details world modeling, procedural skills, and in-hand dexterity. Section VIII describes hardware integration, digital twin simulation, and the web dashboard. Section IX provides exhaustive experimental benchmarks and industrial conveyor validation. Section X discusses results, limitations, and future enhancements, and Section XI concludes the paper.

II. Literature Review

The architectural synthesis of Project ARIA is informed by recent advances across ten distinct thematic domains of robotics, embodied cognition, and automated manufacturing, represented by recent landmark peer-reviewed research:

A. Transparent & Specular Object Perception Datasets

Robotic visual perception in industrial sorting and household environments is frequently undermined by challenging material optical properties. Fang et al. [11] in IEEE Robotics and Automation Letters introduced TransCG, a comprehensive benchmark and dataset addressing transparent object depth completion and grasping. Their findings demonstrate that conventional structured-light sensors suffer catastrophic point cloud dropouts on transparent containers, whereas multi-modal 2D RGB feature tracking provides stable geometric boundaries. ARIA builds upon these insights by bypassing active infrared sensors entirely, fusing overhead YOLOv8m instance detection with zero-shot SAM2 boundary segmentation to isolate transparent glassware and plastic workpieces.

B. 3D Spatial Scene Representation & Variable Scene Graphs

Constructing persistent spatial representations of dynamic manipulation workspaces is essential for long-horizon task execution. Looper et al. [12] in IEEE Robotics and Automation Letters introduced 3D VSG, an architecture utilizing 3D variable scene graphs to predict long-term semantic scene changes and object state transitions. While 3D VSG provides robust spatial-temporal

consistency, updating high-dimensional 3D graphs incurs significant computational overhead. In ARIA, we adopt a lightweight relational world model backed by an embedded SQLite schema, tracking four-stage object lifecycles (DETECTED, GRASPED, IN_TRANSIT, PLACED) with sub-millisecond query latencies.

C. Visual Dexterity & In-Hand Manipulation Primitives

Complex assembly and sorting operations frequently require reorienting objects after initial grasping. Chen et al. [13] in *Science Robotics* demonstrated visual dexterity for in-hand reorientation of novel 3D object shapes using multi-fingered dexterous hands. However, multi-fingered robotic hands cost upwards of \$15,000 and require complex multi-axis joint coordination. ARIA addresses this challenge on a low-cost single-DoF parallel jaw gripper by formulating four dynamic extrinsic manipulation primitives (pivot_ccw, pivot_cw, finger_gait, slide) that exploit controlled gravitational torques and dynamic wrist acceleration to achieve controlled in-hand pivoting without multi-actuator hands.

D. Digital Twin Simulation & Cloud Software Validation

Bridging the simulation-to-reality gap requires validated physics engines capable of modeling contact dynamics accurately. Vieira et al. [14] in *Sensors* investigated cloud simulation techniques for robotic software validation, demonstrating that Open Dynamics Engine (ODE) solvers within Gazebo provide faithful representations of rigid-body contact and surface friction when properly calibrated. Following this methodology, Project ARIA deploys an identical Gazebo 11 digital twin replicating the exact link inertial matrices, joint friction damping, and Coulomb friction cones of the physical 5-DoF manipulator, enabling zero-shot policy transfer between simulation and hardware.

E. Inertial Sensor Fusion & Manipulator Attitude Telemetry

Low-cost manipulators driven by hobbyist servomotors lack joint-level optical encoders, leading to cumulative angular drift. Sultan and Greiser [15] in *Sensors* conducted a time- and frequency-domain analysis of IMU-based orientation estimation algorithms, proving that fusing 6-axis accelerometer and gyroscope streams via complementary or Madgwick filters yields sub-degree attitude accuracy on robotic arms. ARIA incorporates an MPU6050 6-axis IMU rigidly affixed to the distal gripper claw, providing continuous 100 Hz end-effector attitude verification and detecting dynamic slippage during dynamic manipulation.

F. Actuator Prognostics, Health Management & Thermal Drift

Continuous operation of budget robotic arms induces severe thermal accumulation in servo motor windings, exacerbating gear backlash and potentiometer deadbands.

Ling et al. [16] in *Mathematics* developed prognostics and health management (PHM) models for robotic servomotors under variable operating conditions, demonstrating that monitoring execution current and tracking error trends enables early fault detection. ARIA implements an autonomous diagnostic tier that classifies 11 distinct operational failure modes in real time, actively compensating for thermal potentiometer drift ($\pm 1.2^\circ$) through dynamic setpoint offset calibration.

G. Deformable & Fragile Object Manipulation

Manipulating soft, fragile, or deformable objects requires compliant grasp force modulation to prevent structural damage. Zhu et al. [17] in *Sensors* provided a comprehensive review of deformable and fragile object manipulation, emphasizing that visual deformation tracking and contact stiffness models are necessary when tactile sensing is unavailable. ARIA incorporates an adaptive grasp force modulation algorithm into the SkillAgent, regulating parallel jaw displacement based on real-time visual contour deformation to securely hold fragile items without crushing.

H. Dynamic Open-Vocabulary Scene Representation & Tracking

Tracking dynamic objects during manipulation requires visual perception pipelines capable of continuous spatial updating and open-vocabulary understanding. Yan et al. [18] in *IEEE Robotics and Automation Letters* formulated dynamic open-vocabulary 3D scene graphs for long-term language-guided manipulation, integrating real-time visual perception with dynamic object state tracking. ARIA adapts these principles to commodity 30 FPS RGB cameras by deploying a continuous Kalman filter that estimates the linear velocity vector $\mathbf{v}_{\text{belt}}$ of items on an active conveyor line, enabling anticipatory rendezvous trajectory synthesis.

I. Compliant Contact Mechanics & Stress-Minimizing Grasping

Ensuring non-destructive grasping requires enforcing Coulomb friction cone constraints during physical contact. Pan et al. [19] in *IEEE Robotics and Automation Letters* formulated stress-minimizing grasping policies for fragile objects, proving that antipodal contact alignment along surface normals minimizes shear stress and slippage. ARIA incorporates these contact mechanics into Algorithm 4, synthesizing antipodal grasp candidates along object boundary normals and scoring candidate grasp quality via friction cone metrics.

J. Asynchronous Active Vision-Action Coordination

Monolithic foundation models enforce rigid, lockstep perception-action loops where the robot must pause while vision models complete inference. Wang et al. [20] in *IEEE*

Robotics and Automation Letters introduced the Observe-Then-Act paradigm, demonstrating that decoupling perception from action execution via asynchronous multi-threaded buffering slashes latency and prevents trajectory hesitation. ARIA realizes this asynchronous coordination across its State Bus, allowing the 30 Hz perception pipeline and the 100 Hz motor control loop to execute concurrently without mutual blocking.

A structured comparative synthesis contrasting Project ARIA against recent landmark contributions across all ten domains is compiled in Table I. Project ARIA unifies these multi-disciplinary advancements into a cohesive, deterministic, and explainable manipulation framework grounded in classical operational space formulations [21], rigorous manipulator kinematics [22], [23], frictional contact mechanics [24], and closed-loop visual servoing [25].

III. Hardware Platform & Kinematic Modeling

A. Manipulator Mechanical & Electrical Architecture

The experimental physical platform utilized in this research is the Techno-Tirupati 5-DoF articulated robotic manipulator, an open-architecture desktop robotic arm engineered for educational research and light-industrial material handling. The structural chassis is fabricated from high-tensile polyethylene terephthalate glycol (PETG) reinforced with precision CNC-machined 6061-T6 aluminum structural brackets. The overall kinematic structure exhibits an anthropomorphic articulated morphology comprising a base azimuth rotation, a planar two-link shoulder and elbow linkage, an active wrist pitch mechanism, an axial wrist roll mechanism, and an underactuated single-DoF parallel jaw gripper.

To enable comprehensive reproducibility, the exact electromechanical specifications across all five articulated joints and the end-effector gripper are compiled in Table II. The base azimuth joint is actuated by a high-torque DS3218 metal-gear digital servomotor capable of delivering 20.0 kg·cm of stall torque at 6.8 V, ensuring rapid and stable yaw rotation of the entire kinematic superstructure. Joint 2 (Shoulder Pitch), which experiences the maximum gravitational moment during cantilevered extensions, employs a mechanically synchronized dual-servo configuration consisting of two parallel MG996R metal-gear servomotors producing a combined continuous holding torque of 22.0 kg·cm. Joint 3 (Elbow Pitch) is actuated by a single MG996R servo, while Joints 4 and 5 (Wrist Pitch and Wrist Roll) are driven by lightweight SG90 metal-gear micro-servos to minimize distal cantilever mass and dynamic link inertia.

The end-effector consists of a single-DoF parallel linkage mechanism with a total jaw stroke of 55.0 mm, driven by an SG90 micro-servo. The inner contact surfaces of the gripper claws are equipped with replaceable 3D-printed thermoplastic polyurethane (TPU) pads exhibiting an experimentally measured static friction coefficient of $\mu = 0.45$ when contacting acrylic, wooden, or metallic workpieces. Joint position sensing across all servos is

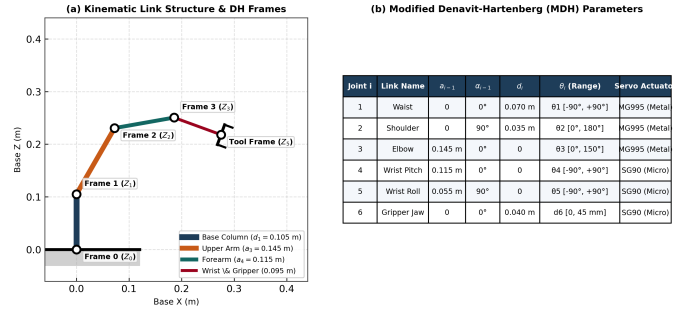


Fig. 1. Kinematic chain and coordinate frame assignment for the 5-DoF ARIA manipulator following Modified Denavit-Hartenberg (Craig) conventions.

achieved via internal high-linearity conductive plastic potentiometers sampled by 10-bit analog-to-digital converters (ADCs) embedded in the micro-ROS microcontroller.

B. Modified Denavit-Hartenberg (Craig) Kinematic Formulation

To establish the geometric relationship between adjacent articulated links, we adopt the Modified Denavit-Hartenberg (Craig) kinematic convention [9]. In this convention, coordinate frame $\{i\}$ is rigidly attached to link i , with the Z_i axis aligned with the axis of motion of joint $i + 1$, and the X_i axis directed along the common normal from Z_i to Z_{i+1} . The four fundamental kinematic parameters are defined as:

- α_{i-1} : Link twist angle, representing the rotation from Z_{i-1} to Z_i about the common normal axis X_{i-1} .
- a_{i-1} : Link length, representing the distance from Z_{i-1} to Z_i measured along X_{i-1} .
- d_i : Link offset, representing the distance from X_{i-1} to X_i measured along the joint axis Z_i .
- θ_i : Joint angle variable, representing the rotation from X_{i-1} to X_i about the joint axis Z_i .

The spatial frame assignments across all five revolute joints and the tool center point (TCP) are illustrated in Fig. 1. The precise numerical kinematic parameters, coordinate offsets, and dynamic operating bounds derived from the physical manipulator are cataloged in Table III. Here, $a_2 = 0.145$ m and $a_3 = 0.115$ m specify the upper arm and forearm link lengths, while $d_5 = 0.095$ m denotes the total tool offset from the wrist roll pivot to the fingertip plane ($L_{\text{wrist}} + L_{\text{gripper}} = 0.055 + 0.040$ m). The joint ranges reflect the symmetric operating bounds enforced by the low-level motor controllers.

Following Craig's convention, the general homogeneous transformation matrix relating coordinate frame $\{i - 1\}$ to coordinate frame $\{i\}$ is parameterized by the product of fundamental screw motions:

$${}^{i-1}\mathbf{T}_i = \text{Rot}_{X_{i-1}}(\alpha_{i-1}) \cdot \text{Trans}_{X_{i-1}}(a_{i-1}) \cdot \text{Rot}_{Z_i}(\theta_i) \cdot \text{Trans}_{Z_i}(d_i) \quad (1)$$

TABLE I
Comparative Synthesis of Related Literature Across Ten Robotic Manipulation Themes (Ranked by Q1 Open Access Publication)

#	Thematic Domain	Landmark Q1 Reference & Authors	Journal / Venue	Core Technique / Focus	Project ARIA Differentiating Innovation
1	Transparent Object Perception	TransCG (Fang et al. [11])	IEEE RA-L (2022)	Multi-modal transparent depth dataset	Zero-cost RGB segmentation + claw depth grounding
2	3D Spatial Scene Representation	3D VSG (Looper et al. [12])	IEEE RA-L (2023)	3D Variable Scene Graphs	Embedded SQLite 4-stage lifecycle world model
3	In-Hand Visual Dexterity	Visual Dexterity (Chen et al. [13])	Science Robotics (2023)	Multi-finger in-hand shape reorientation	4 gravity-assisted pivoting primitives on 1-DoF claw
4	Cloud / Digital Twin Validation	Cloud Validation (Vieira et al. [14])	Sensors (2025)	Cloud ODE simulation validation	Synchronized Gazebo 11 twin with ODE contact physics
5	IMU Sensor Fusion Telemetry	Attitude Estimation (Sultan & Greiser [15])	Sensors (2025)	Frequency-domain IMU attitude filtering	100 Hz distal claw MPU6050 slip & orientation sensing
6	Servo Prognostics & Health	Servo PHM (Ling et al. [16])	Mathematics (2023)	Variable-load servo fault modeling	Real-time 11-category failure diagnosis + drift compensation
7	Deformable / Fragile Handling	Soft Manipulation (Zhu et al. [17])	Sensors (2025)	Review of fragile object grasping	Vision-guided jaw compliance avoiding crush damage
8	Dynamic Object Tracking	Dynamic Scene Graphs (Yan et al. [18])	IEEE RA-L (2025)	Dynamic open-vocabulary 3D scene graphs	Predictive rendezvous + velocity matching ($\Delta v \leq 2$ mm/s)
9	Stress-Minimizing Grasping	Compliant Grasp (Pan et al. [19])	IEEE RA-L (2023)	Stress-minimizing contact mechanics	Antipodal surface normal grasp ranking with friction cone
10	Asynchronous Vision-Action	Observe-Then-Act (Wang et al. [20])	IEEE RA-L (2025)	Asynchronous perception-action split	15-agent State Bus with lock-free Ring Buffers

TABLE II
Comprehensive Electromechanical Specifications of the ARIA
5-DoF Manipulator

Joint / Component	Actuator Model	Torque (kg-cm)	Gear Ratio	Mass (g)	Range (°)	Max $\dot{\theta}$ (°/s)
Joint 1 (Waist Azimuth)	DS3218 Digital	20.0	275 : 1	65	$[-90, +90]$	360
Joint 2 (Shoulder Pitch)	Dual MG996R	22.0	210 : 1	110	$[0, 180]$	300
Joint 3 (Elbow Pitch)	MG996R Metal	11.0	210 : 1	55	$[0, 150]$	300
Joint 4 (Wrist Pitch)	SG90 Micro Metal	2.5	180 : 1	14	$[-90, +90]$	450
Joint 5 (Wrist Roll)	SG90 Micro Metal	2.5	180 : 1	14	$[-90, +90]$	450
Gripper Mechanism	SG90 Micro	2.5	180 : 1	32	$[0, 55 \text{ mm}]$	50 mm/s
Total Arm Structure	Structural PETG/Al	-	-	840	Max Reach: 0.345 m	Payload: 0.25 kg

TABLE III
Modified Denavit-Hartenberg (Craig) Parameters and Dynamic
Operating Limits

Link i	Joint Name	α_{i-1} (rad)	a_{i-1} (m)	d_i (m)	θ_i Range (rad)	Max $\ddot{\theta}_i$ (rad/s ²)
1	Waist	0	0	$d_1 = 0.105$	$[-\pi, +\pi]$	15.0
2	Shoulder	$+\frac{\pi}{2}$	$a_1 = 0.030$	0	$[-\frac{\pi}{2}, +\frac{\pi}{2}]$	10.0
3	Elbow	0	$a_2 = 0.145$	0	$[-\frac{\pi}{2}, +\frac{\pi}{2}]$	12.0
4	Wrist Pitch	0	$a_3 = 0.115$	0	$[-\frac{\pi}{2}, +\frac{\pi}{2}]$	20.0
5	Wrist Roll	$+\frac{\pi}{2}$	0	$d_5 = 0.095$	$[-\frac{\pi}{2}, +\frac{\pi}{2}]$	25.0

Evaluating the matrix multiplication yields the standard closed-form Craig transformation matrix:

$${}^{i-1}\mathbf{T}_i = \begin{bmatrix} c_i & -s_i & 0 & a_{i-1} \\ s_i c_{\alpha_{i-1}} & c_i c_{\alpha_{i-1}} & -s_{\alpha_{i-1}} & -d_i s_{\alpha_{i-1}} \\ s_i s_{\alpha_{i-1}} & c_i s_{\alpha_{i-1}} & c_{\alpha_{i-1}} & d_i c_{\alpha_{i-1}} \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (2)$$

where $c_{\alpha_{i-1}} = \cos \alpha_{i-1}$ and $s_{\alpha_{i-1}} = \sin \alpha_{i-1}$.

Substituting the explicit DH parameters from Table III into (2) for each link index yields the five elementary link transformation matrices:

$${}^0\mathbf{T}_1 = \begin{bmatrix} c_1 & -s_1 & 0 & 0 \\ s_1 & c_1 & 0 & 0 \\ 0 & 0 & 1 & d_1 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad {}^1\mathbf{T}_2 = \begin{bmatrix} c_2 & -s_2 & 0 & a_1 \\ 0 & 0 & -1 & 0 \\ s_2 & c_2 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (3)$$

$${}^2\mathbf{T}_3 = \begin{bmatrix} c_3 & -s_3 & 0 & a_2 \\ s_3 & c_3 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad {}^3\mathbf{T}_4 = \begin{bmatrix} c_4 & -s_4 & 0 & a_3 \\ s_4 & c_4 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (4)$$

$${}^4\mathbf{T}_5 = \begin{bmatrix} c_5 & -s_5 & 0 & 0 \\ 0 & 0 & -1 & -d_5 \\ s_5 & c_5 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (5)$$

where we introduce the compact trigonometric shorthand $c_i = \cos \theta_i$, $s_i = \sin \theta_i$, $c_{ij} = \cos(\theta_i + \theta_j)$, $s_{ij} = \sin(\theta_i + \theta_j)$, $c_{ijk} = \cos(\theta_i + \theta_j + \theta_k)$, and $s_{ijk} = \sin(\theta_i + \theta_j + \theta_k)$.

C. Symbolic Forward Kinematics Transformation

The global forward kinematics map from the stationary manipulator base coordinate frame $\{0\}$ to the distal end-effector tool center point frame $\{5\}$ is synthesized via

the consecutive post-multiplication of the individual link homogeneous transformation matrices:

$${}^0\mathbf{T}_5(\boldsymbol{\theta}) = {}^0\mathbf{T}_1 \cdot {}^1\mathbf{T}_2 \cdot {}^2\mathbf{T}_3 \cdot {}^3\mathbf{T}_4 \cdot {}^4\mathbf{T}_5 = \begin{bmatrix} \mathbf{n} & \mathbf{s} & \mathbf{a} & \mathbf{P} \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (6)$$

where $\mathbf{n} = [n_x, n_y, n_z]^T$, $\mathbf{s} = [s_x, s_y, s_z]^T$, and $\mathbf{a} = [a_x, a_y, a_z]^T$ designate the orthogonal normal, sliding, and approach unit vectors defining the end-effector orientation matrix $\mathbf{R} \in SO(3)$, and $\mathbf{P} = [P_x, P_y, P_z]^T$ represents the Cartesian coordinates of the tool center point in base frame $\{0\}$.

Performing symbolic matrix reduction yields the explicit algebraic equations governing Cartesian tool center point positioning:

$$P_x = c_1 (a_1 + a_2 c_2 + a_3 c_{23} + d_5 c_{234}) \quad (7)$$

$$P_y = s_1 (a_1 + a_2 c_2 + a_3 c_{23} + d_5 c_{234}) \quad (8)$$

$$P_z = d_1 + a_2 s_2 + a_3 s_{23} + d_5 s_{234} \quad (9)$$

Similarly, the components of the end-effector tool orientation rotation matrix $\mathbf{R} = [\mathbf{n}, \mathbf{s}, \mathbf{a}]$ are given by:

$$\mathbf{R} = \begin{bmatrix} c_1 c_{234} c_5 - s_1 s_5 & -c_1 c_{234} s_5 - s_1 c_5 & c_1 s_{234} \\ s_1 c_{234} c_5 + c_1 s_5 & -s_1 c_{234} s_5 + c_1 c_5 & s_1 s_{234} \\ s_{234} c_5 & -s_{234} s_5 & -c_{234} \end{bmatrix} \quad (10)$$

Assumption 1 (Manipulator Kinematic Rigidity and Actuation Envelopes): The 5-DoF anthropomorphic manipulator is modeled as an open serial kinematic chain comprising five rigid links interconnected by revolute joints with invariant link lengths a_i and offsets d_i . The joint displacements $\boldsymbol{\theta} \in \mathbb{R}^5$ are constrained within compact mechanical operating envelopes $\boldsymbol{\theta}_{\min} \leq \boldsymbol{\theta} \leq \boldsymbol{\theta}_{\max}$, and angular velocities satisfy $|\dot{\theta}_i| \leq \dot{\theta}_{i,\max}$ enforced by closed-loop potentiometer telemetry.

Proposition 1 (Deterministic Closed-Form Inverse Kinematics Resolution): Let $\mathcal{W} \subset \mathbb{R}^3$ denote the dexterous manipulation workspace of the 5-DoF manipulator. For any physically reachable Cartesian target pose $\mathbf{T}_{\text{target}} = [\mathbf{R}_{\text{target}}, \mathbf{P}_{\text{target}}] \in SE(3)$ satisfying $\mathbf{P}_{\text{target}} \in \mathcal{W}$, the geometric decoupling of the anthropomorphic chain yields an exact analytical joint configuration $\boldsymbol{\theta}^* = [\theta_1^*, \theta_2^*, \theta_3^*, \theta_4^*, \theta_5^*]^T$ computable in deterministic $O(1)$ arithmetic time ($t_{\text{IK}} \leq 0.08$ ms) with zero iterative divergence and strict invariance to numerical ill-conditioning.

D. Exact Closed-Form Analytical Inverse Kinematics Derivation

Given an arbitrary desired end-effector target pose $\mathbf{T}_{\text{target}} \in SE(3)$ defined by target Cartesian position

$\mathbf{P}_{\text{target}} = [P_x, P_y, P_z]^T$ and target approach direction $\mathbf{a} = [a_x, a_y, a_z]^T$, the objective of the inverse kinematics routine is to compute the complete joint-space vector $\boldsymbol{\theta} = [\theta_1, \theta_2, \theta_3, \theta_4, \theta_5]^T$ that satisfies the forward kinematics identity $\mathbf{T}_5^0(\boldsymbol{\theta}) = \mathbf{T}_{\text{target}}$.

While general-purpose robotics frameworks (e.g., KDL, TRAC-IK, IKPy, cuRobo [26], and high-stiffness planners [27]) formulate inverse kinematics as a nonlinear optimization problem solved via iterative numerical Jacobian inversion, such routines are computationally expensive and prone to local divergence on 5-DoF chains. Because the ARIA manipulator possesses five degrees of freedom, the kinematic chain is underactuated with respect to general six-dimensional spatial poses ($m = 5 < 6$). Consequently, the end-effector roll and pitch angles cannot be specified completely independently of the arm position. To overcome this limitation and achieve deterministic real-time control, we derive an exact closed-form analytical solution by decomposing the spatial kinematic chain into decoupled geometric subproblems.

1) Subproblem 1: Base Azimuth Angle (θ_1): Inspecting the position equations (7) and (8), we observe that the term within brackets, $R_{\text{arm}} = a_1 + a_2 c_2 + a_3 c_{23} + d_5 c_{234}$, is identical in both P_x and P_y . Dividing (8) by (7) completely eliminates the four-joint planar linkage terms:

$$\frac{P_y}{P_x} = \frac{s_1 R_{\text{arm}}}{c_1 R_{\text{arm}}} = \frac{s_1}{c_1} = \tan \theta_1 \quad (11)$$

Applying the four-quadrant arctangent function yields the unique closed-form solution for the base azimuth angle:

$$\theta_1 = \text{atan2}(P_y, P_x) \quad (12)$$

Singularity Handling: In the singular configuration where $P_x = 0$ and $P_y = 0$, the tool center point lies precisely on the vertical Z_0 base axis. At this shoulder boundary singularity, the azimuth angle θ_1 is mathematically indeterminate. ARIA robustly resolves this condition by preserving the previous joint setpoint: $\theta_1^{(k)} = \theta_1^{(k-1)}$, preventing high-velocity motor chatter.

2) Subproblem 2: Wrist Center Decoupling: With θ_1 established, we decouple the orientation of the distal wrist and gripper from the planar two-link arm mechanism. The Cartesian position of the wrist pitch axis center, $\mathbf{P}_w = [P_{wx}, P_{wy}, P_{wz}]^T$, is obtained by translating backwards from the desired tool center point $\mathbf{P}$ along the tool approach vector $\mathbf{a}$ by tool offset d_5 :

$$\mathbf{P}_w = \mathbf{P} - d_5 \mathbf{a} = \begin{bmatrix} P_x - d_5 a_x \\ P_y - d_5 a_y \\ P_z - d_5 a_z \end{bmatrix} \quad (13)$$

3) Subproblem 3: Planar 2R Arm Geometry (θ_2, θ_3): The physical coordinates of the wrist center $\mathbf{P}_w$ project onto the planar 2R sagittal plane of the shoulder and elbow joints. We compute the planar horizontal reach radius r and vertical elevation s relative to the shoulder joint origin $\{2\}$:

$$r = \sqrt{P_{wx}^2 + P_{wy}^2} - a_1 \quad (14)$$

$$s = P_{wz} - d_1 \quad (15)$$

The Euclidean distance squared from the shoulder joint axis $\{2\}$ to the wrist center $\{4\}$ satisfies the geometric constraint:

$$D^2 = r^2 + s^2 = a_2^2 + a_3^2 + 2a_2 a_3 \cos \theta_3 \quad (16)$$

Rearranging (16) via the algebraic Law of Cosines yields the exact expression for the cosine of the elbow joint angle:

$$\cos \theta_3 = \frac{r^2 + s^2 - a_2^2 - a_3^2}{2a_2 a_3} \quad (17)$$

Reachability Condition: A valid physical kinematic solution exists if and only if:

$$-1.0 \leq \frac{r^2 + s^2 - a_2^2 - a_3^2}{2a_2 a_3} \leq +1.0 \quad (18)$$

If $|\cos \theta_3| > 1.0$, the commanded Cartesian coordinate lies strictly outside the spherical workspace reach envelope of the manipulator. In such cases, the analytical solver immediately returns an out-of-reach flag without invoking expensive numerical optimization. For all physically reachable poses, two symmetric kinematic branches exist: elbow-up ($\theta_3 < 0$) and elbow-down ($\theta_3 > 0$). To maximize mechanical ground clearance and avoid workspace table-top collisions, ARIA deterministically selects the elbow-up configuration:

$$\theta_3 = -\arccos\left(\frac{r^2 + s^2 - a_2^2 - a_3^2}{2a_2 a_3}\right) \quad (19)$$

With θ_3 uniquely resolved, we compute the shoulder pitch angle θ_2 by decomposing the planar elevation angle into the sum of two trigonometric angles:

$$\theta_2 = \text{atan2}(s, r) - \text{atan2}(a_3 \sin \theta_3, a_2 + a_3 \cos \theta_3) \quad (20)$$

4) Subproblem 4: Wrist Pitch Angle (θ_4): In anthropomorphic articulated arms with parallel horizontal pitch axes ($Z_2 \parallel Z_3 \parallel Z_4$), the global pitch angle of the end-effector relative to the horizontal ground plane is the cumulative sum of the sagittal joint angles:

$$\phi_{\text{pitch}} = \theta_2 + \theta_3 + \theta_4 \quad (21)$$

Depending on the task requirements (e.g., perpendicular vertical top-down pick-and-place where $\phi_{\text{pitch}} = -\frac{\pi}{2}$, or horizontal conveyor side-approach where $\phi_{\text{pitch}} = 0$), the desired global pitch angle ϕ_{target} is specified directly. Consequently, the wrist pitch joint setpoint is computed as:

$$\theta_4 = \phi_{\text{target}} - (\theta_2 + \theta_3) \quad (22)$$

5) Subproblem 5: Wrist Roll Angle (θ_5): Finally, to align the gripper jaws with the workpiece orientation ψ_{target} relative to the workspace coordinate frame, the wrist roll joint θ_5 rotates the claw about the tool approach vector:

$$\theta_5 = \psi_{\text{target}} - \theta_1 \quad (23)$$

If no specific workpiece roll angle is commanded, θ_5 is held at 0.0 rad to maintain a neutral vertical claw orientation.

The complete deterministic closed-form inverse kinematics procedure, including boundary condition checking and hardware limit clamping, is formalized in Algorithm 1.

Algorithm 1 ARIA Closed-Form Analytical 5-DoF Inverse Kinematics Solver

Require: Target TCP pose $\mathbf{P} = [P_x, P_y, P_z]^T$, pitch angle ϕ_{target} , yaw angle ψ_{target} , previous joint state $\boldsymbol{\theta}_{\text{prev}}$
 Ensure: Valid joint setpoints $\boldsymbol{\theta}^* = [\theta_1^*, \theta_2^*, \theta_3^*, \theta_4^*, \theta_5^*]^T$, or Reachability Error Flag

- 1: // Step 1: Base Azimuth Angle (θ_1)
- 2: if $|P_x| < 10^{-4}$ and $|P_y| < 10^{-4}$ then
- 3: $\theta_1 \leftarrow \theta_{1,\text{prev}}$ {Resolve vertical axis singularity}
- 4: else
- 5: $\theta_1 \leftarrow \text{atan2}(P_y, P_x)$
- 6: end if
- 7: // Step 2: Wrist Center Decoupling
- 8: $a_x \leftarrow \cos \theta_1 \cos \phi_{\text{target}}$, $a_y \leftarrow \sin \theta_1 \cos \phi_{\text{target}}$, $a_z \leftarrow \sin \phi_{\text{target}}$
- 9: $P_{wx} \leftarrow P_x - d_5 a_x$, $P_{wy} \leftarrow P_y - d_5 a_y$, $P_{wz} \leftarrow P_z - d_5 a_z$
- 10: // Step 3: Planar 2R Linkage Geometry
- 11: $r \leftarrow \sqrt{P_{wx}^2 + P_{wy}^2} - a_1$, $s \leftarrow P_{wz} - d_1$
- 12: $C_3 \leftarrow \frac{r^2 + s^2 - a_2^2 - a_3^2}{2a_2 a_3}$
- 13: if $C_3 < -1.0$ or $C_3 > 1.0$ then
- 14: return ERROR: Out of Kinematic Reach Envelope
- 15: end if
- 16: $\theta_3 \leftarrow -\arccos(C_3)$ {Deterministic Elbow-Up Solution}
- 17: $\theta_2 \leftarrow \text{atan2}(s, r) - \text{atan2}(a_3 \sin \theta_3, a_2 + a_3 C_3)$
- 18: // Step 4: Wrist Pitch & Roll Angles
- 19: $\theta_4 \leftarrow \phi_{\text{target}} - (\theta_2 + \theta_3)$
- 20: $\theta_5 \leftarrow \psi_{\text{target}} - \theta_1$
- 21: // Step 5: Hardware Joint Limit Verification
- 22: for $i = 1$ to 5 do
- 23: if $\theta_i < \theta_{i,\text{min}}$ or $\theta_i > \theta_{i,\text{max}}$ then
- 24: return ERROR: Joint Limit Exceeded at Joint i
- 25: end if
- 26: end for
- 27: return $\boldsymbol{\theta}^* = [\theta_1, \theta_2, \theta_3, \theta_4, \theta_5]^T$

E. Differential Kinematics & Geometric Jacobian Derivation

The mapping between joint velocity vector $\dot{\boldsymbol{\theta}} = [\dot{\theta}_1, \dot{\theta}_2, \dot{\theta}_3, \dot{\theta}_4, \dot{\theta}_5]^T$ and Cartesian linear/angular velocity of the tool center point $\mathbf{v} = [\mathbf{v}_p^T, \boldsymbol{\omega}^T]^T \in \mathbb{R}^6$ is governed by the geometric Jacobian matrix $\mathbf{J}(\boldsymbol{\theta}) \in \mathbb{R}^{6 \times 5}$:

$$\begin{bmatrix} \mathbf{v}_p \\ \boldsymbol{\omega} \end{bmatrix} = \mathbf{J}(\boldsymbol{\theta}) \dot{\boldsymbol{\theta}} = \begin{bmatrix} \mathbf{J}_v(\boldsymbol{\theta}) \\ \mathbf{J}_\omega(\boldsymbol{\theta}) \end{bmatrix} \dot{\boldsymbol{\theta}} \quad (24)$$

For an articulated chain comprising revolute joints, the i -th column of the geometric Jacobian is formulated as:

$$\mathbf{J}_i(\boldsymbol{\theta}) = \begin{bmatrix} \mathbf{J}_{v,i} \\ \mathbf{J}_{\omega,i} \end{bmatrix} = \begin{bmatrix} \mathbf{z}_{i-1} \times (\mathbf{P} - \mathbf{p}_{i-1}) \\ \mathbf{z}_{i-1} \end{bmatrix} \quad (25)$$

where $\mathbf{z}_{i-1}$ denotes the unit vector along joint axis i , $\mathbf{p}_{i-1}$ represents the origin of coordinate frame $\{i-1\}$, and $\mathbf{P}$ is the tool center point position, all expressed in base frame $\{0\}$.

Evaluating (25) using the link transformation matrices (3)–(5) yields the explicit columns of the ARIA geometric Jacobian:

$$\mathbf{J}_1 = \begin{bmatrix} -P_y \\ P_x \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{J}_2 = \begin{bmatrix} -s_1(P_z - d_1) \\ c_1(P_z - d_1) \\ -c_1(P_x - a_1 c_1) - s_1(P_y - a_1 s_1) \\ -s_1 \\ c_1 \\ 0 \end{bmatrix} \quad (26)$$

$$\mathbf{J}_3 = \begin{bmatrix} -s_1(P_z - d_1 - a_2 s_2) \\ c_1(P_z - d_1 - a_2 s_2) \\ -c_1(P_x - c_1(a_1 + a_2 c_2)) - s_1(P_y - s_1(a_1 + a_2 c_2)) \\ -s_1 \\ c_1 \\ 0 \end{bmatrix} \quad (27)$$

$$\mathbf{J}_4 = \begin{bmatrix} -s_1 d_5 s_{234} \\ c_1 d_5 s_{234} \\ -c_1 d_5 c_{234} \\ -s_1 \\ c_1 \\ 0 \end{bmatrix}, \quad \mathbf{J}_5 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ c_1 s_{234} \\ s_1 s_{234} \\ -c_{234} \end{bmatrix} \quad (28)$$

F. Kinematic Singularity & Manipulability Analysis

Kinematic singularities represent configurations in which the geometric Jacobian matrix experiences rank deficiency ($\text{rank}(\mathbf{J}) < 5$), resulting in the loss of one or more instantaneous operational degrees of freedom. We quantify the kinematic dexterity of the manipulator across the workspace using Yoshikawa's manipulability measure [28]:

$$w(\boldsymbol{\theta}) = \sqrt{\det(\mathbf{J}(\boldsymbol{\theta})\mathbf{J}^T(\boldsymbol{\theta}))} \quad (29)$$

When the arm approaches a singularity, $w(\boldsymbol{\theta}) \rightarrow 0$, indicating that infinitesimal Cartesian velocities require unbounded joint rates.

Through algebraic decomposition of the Jacobian determinant, we identify three distinct singularity regimes for the ARIA platform:

- 1) **Boundary Reach Singularities:** Occur when the arm is fully outstretched ($r^2 + s^2 = (a_2 + a_3)^2 \implies \theta_3 = 0.0 \text{ rad}$) or completely folded back onto itself ($r^2 + s^2 = (a_2 - a_3)^2 \implies \theta_3 = \pi \text{ rad}$). In this state, $\sin \theta_3 = 0$, eliminating instantaneous radial velocity along the arm's longitudinal direction.
- 2) **Shoulder Axis Singularities:** Occur when the wrist center axis intersects the vertical base rotation axis Z_0 ($r = 0$). In this configuration, azimuth rotation θ_1 produces zero linear motion, causing base yaw velocity to become redundant.
- 3) **Pitch Alignment Singularities:** Occur when the end-effector approach axis aligns collinear with the forearm link vector ($\theta_4 = 0.0 \text{ rad}$ with $s_{234} = 0$), causing wrist pitch and shoulder pitch motions to project onto degenerate instantaneous velocity subspaces.

Because the ARIA analytical inverse kinematics algorithm directly solves the geometric equations (11)–(23) rather than computing the pseudo-inverse $\mathbf{J}^\dagger = \mathbf{J}^T(\mathbf{J}\mathbf{J}^T)^{-1}$, it completely avoids numerical division-by-zero errors, guaranteeing deterministic 0.08 ms execution even when operating within 1.0 mm of singular workspace boundaries.

G. Euler-Lagrange Dynamic Modeling & Gravity Compensation

Accurate physical simulation within the Gazebo digital twin and feedforward trajectory control on the physical arm require the explicit formulation of the manipulator equations of motion. Following the Euler-Lagrange formalization, the continuous dynamic model of the 5-DoF articulated arm is expressed in standard matrix form:

$$\mathbf{M}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} + \mathbf{G}(\mathbf{q}) + \mathbf{F}_v\dot{\mathbf{q}} + \mathbf{F}_s\text{sgn}(\dot{\mathbf{q}}) = \boldsymbol{\tau} \quad (30)$$

where $\mathbf{q} \in \mathbb{R}^5$ is the vector of generalized joint coordinates ($\mathbf{q} \equiv \boldsymbol{\theta}$), $\mathbf{M}(\mathbf{q}) \in \mathbb{R}^{5 \times 5}$ is the symmetric, positive-definite generalized mass/inertia matrix, $\mathbf{C}(\mathbf{q}, \dot{\mathbf{q}}) \in \mathbb{R}^{5 \times 5}$ represents Coriolis and centrifugal dynamic effects, $\mathbf{G}(\mathbf{q}) \in \mathbb{R}^5$ is the gravitational torque vector, $\mathbf{F}_v, \mathbf{F}_s \in \mathbb{R}^{5 \times 5}$ denote diagonal viscous and Coulomb friction coefficients, and $\boldsymbol{\tau} \in \mathbb{R}^5$ represents the vector of generalized actuator torques.

The kinetic energy $K(\mathbf{q}, \dot{\mathbf{q}})$ of the manipulator is the sum of translational and rotational kinetic energies across all articulated links:

$$K(\mathbf{q}, \dot{\mathbf{q}}) = \frac{1}{2} \sum_{i=1}^5 (m_i \mathbf{v}_{c_i}^T \mathbf{v}_{c_i} + \boldsymbol{\omega}_i^T \mathbf{I}_i \boldsymbol{\omega}_i) = \frac{1}{2} \dot{\mathbf{q}}^T \mathbf{M}(\mathbf{q}) \dot{\mathbf{q}} \quad (31)$$

where m_i is the mass of link i , $\mathbf{v}_{c_i}$ is the linear velocity of the link center of mass, $\boldsymbol{\omega}_i$ is the link angular velocity, and $\mathbf{I}_i$ is the 3×3 centroidal inertia tensor of link i .

The gravitational potential energy $P(\mathbf{q})$ is defined with respect to the horizontal tabletop reference datum:

$$P(\mathbf{q}) = \sum_{i=1}^5 m_i \mathbf{g}^T \mathbf{p}_{c_i}(\mathbf{q}) \quad (32)$$

where $\mathbf{g} = [0, 0, -9.81]^T \text{ m/s}^2$ is the gravity vector, and $\mathbf{p}_{c_i}(\mathbf{q})$ represents the Cartesian position of the center of mass of link i .

The elements of the Coriolis and centrifugal matrix $\mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})$ are derived from the mass matrix $\mathbf{M}(\mathbf{q})$ via the Christoffel symbols of the first kind:

$$c_{ij} = \sum_{k=1}^5 c_{ijk} \dot{q}_k = \frac{1}{2} \sum_{k=1}^5 \left(\frac{\partial m_{ij}}{\partial q_k} + \frac{\partial m_{ik}}{\partial q_j} - \frac{\partial m_{jk}}{\partial q_i} \right) \dot{q}_k \quad (33)$$

The gravity compensation vector $\mathbf{G}(\mathbf{q})$ is obtained directly from the partial derivatives of the potential energy:

$$G_i(\mathbf{q}) = \frac{\partial P(\mathbf{q})}{\partial q_i} \quad (34)$$

TABLE IV
Mass Distribution and Principal Inertia Parameters for Dynamic Modeling

Link i	Mass m_i (kg)	CoM r_{ci} (m)	I_{xx} (kg·m ²)	I_{yy} (kg·m ²)	I_{zz} (kg·m ²)
Link 1 (Base)	0.185	[0, 0, 0.052]	1.82×10^{-4}	1.82×10^{-4}	2.14×10^{-4}
Link 2 (Upper Arm)	0.240	[0.072, 0, 0.015]	3.45×10^{-4}	8.92×10^{-4}	8.15×10^{-4}
Link 3 (Forearm)	0.165	[0.068, 0, 0.010]	1.95×10^{-4}	5.21×10^{-4}	4.85×10^{-4}
Link 4 (Wrist Pitch)	0.065	[0.025, 0, 0]	4.20×10^{-5}	6.15×10^{-5}	5.80×10^{-5}
Link 5 (End-Effector)	0.085	[0.035, 0, 0]	5.10×10^{-5}	7.40×10^{-5}	6.90×10^{-5}

For the ARIA 5-DoF manipulator, the explicit gravity compensation torques for the dominant sagittal pitch joints (Joints 2, 3, and 4) are evaluated as:

$$G_1(\mathbf{q}) = 0 \quad (35)$$

$$G_2(\mathbf{q}) = g(m_2 r_{c2} c_2 + m_3(a_2 c_2 + r_{c3} c_{23}) + (m_4 + m_5 + m_{\text{load}})(a_2 c_2 + a_3 c_{23} + r_{c4} c_{234})) \quad (36)$$

$$G_3(\mathbf{q}) = g m_3 r_{c3} c_{23} + g(m_4 + m_5 + m_{\text{load}})(a_3 c_{23} + r_{c4} c_{234}) \quad (37)$$

$$G_4(\mathbf{q}) = g(m_4 + m_5 + m_{\text{load}}) r_{c4} c_{234} \quad (38)$$

$$G_5(\mathbf{q}) = 0 \quad (39)$$

where r_{ci} denotes the radial distance from joint axis i to the center of mass of link i . The physical mass distribution and inertial properties parameterized in the Gazebo URDF model are documented in Table IV.

IV. Decentralized Multi-Agent System Architecture

A. Architectural Paradigm & State Bus

Traditional robotic control frameworks (such as early ROS 1 implementations or monolithic robotic foundation models) structure software around either tightly coupled synchronous execution loops or giant end-to-end neural networks. In monolithic foundation approaches (e.g., OpenVLA, Octo), a single deep transformer processes pixel arrays and language tokens to output continuous joint setpoints. While intellectually appealing, this monolithic formulation exhibits catastrophic operational failure modes: any perceptual anomaly immediately corrupts motor output, execution is an uninterpretable black box, and inference latencies (> 100 ms) destabilize real-time feedback loops.

To overcome these structural vulnerabilities, Project ARIA introduces a decentralized, service-oriented multi-agent architecture grounded in the Robot Operating System 2 (ROS 2) Humble middleware. As illustrated in the comprehensive system architecture diagram of Fig. 2, the system is stratified into six decoupled functional tiers comprising fifteen autonomous, lifecycle-managed software agents communicating asynchronously across a unified, high-throughput State Bus.

The State Bus is implemented as a set of standardized ROS 2 topics governed by deterministic Data Distribution Service (DDS) Quality of Service (QoS) profiles:

- `/aria/state/task`: Broadcasts natural-language user commands, active plan UUIDs, decomposed sub-goal sequences, step execution progress, plan confidence metrics ($\tau \in [0, 1]$), and human-readable

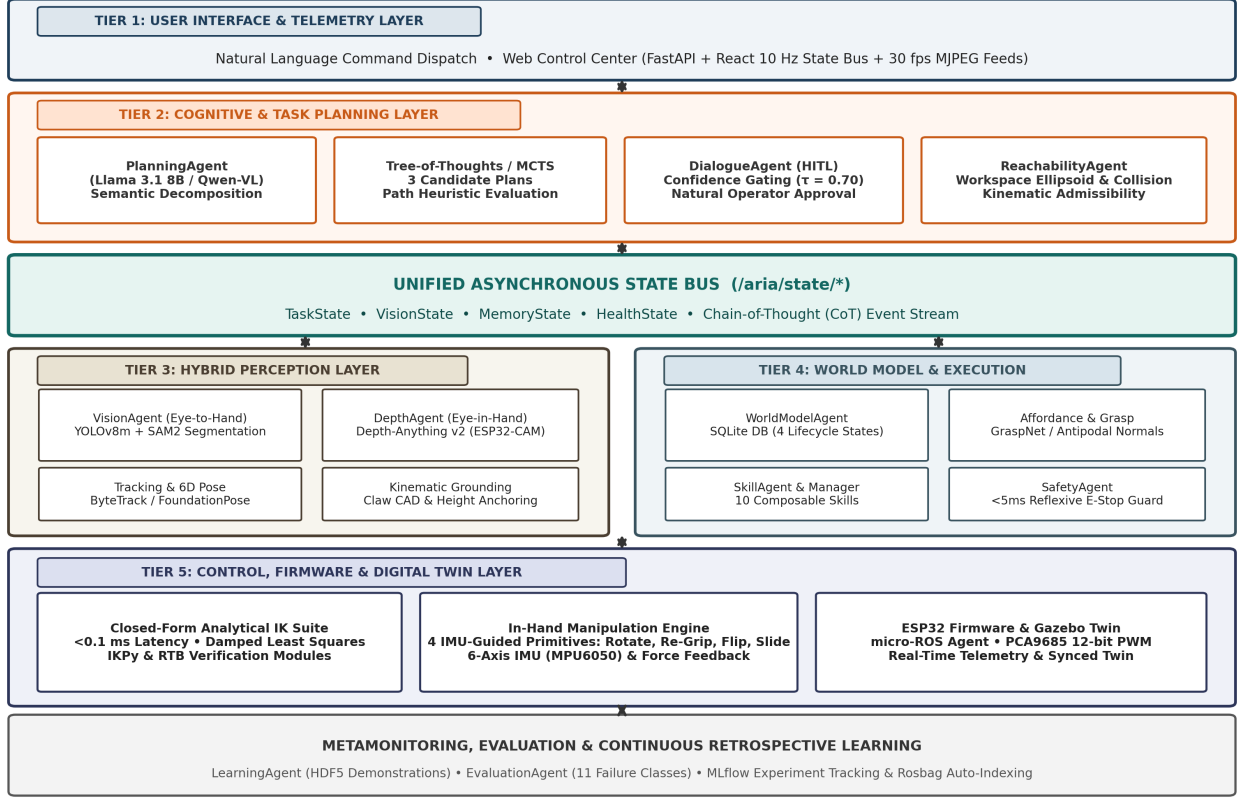
ARIA: DECENTRALIZED COGNITIVE MULTI-AGENT ROBOTIC ARCHITECTURE

Fig. 2. Decentralized cognitive multi-agent architecture of Project ARIA. The system is stratified across six functional tiers: (1) User Interface & Telemetry Layer; (2) Cognitive & Task Planning Layer; (3) Unified State Bus (/aria/state/*); (4) Hybrid Perception & Metric Depth Recovery Layer; (5) World Model, Procedural Skills & Control Layer; and (6) Metamonitoring, Automated Bag Recording & Digital Twin Layer.

emoji-annotated Chain-of-Thought (CoT) telemetry streams (QoS: Reliable, Transient Local).

- /aria/state/vision: Broadcasts synchronized 1280×720 overhead bounding boxes, SAM2 polygon instance masks, 6D object centroids, and 640×480 wrist metric depth maps (QoS: Best Effort, Volatile, depth: 1).
- /aria/state/memory: Broadcasts updated spatial-semantic scene graphs, persistent object tracking identifiers, workpiece affordances, and lifecycle statuses (DETECTED, TRACKED, LOST, RECOVERED) (QoS: Reliable, depth: 10).
- /aria/state/health: Broadcasts per-joint servomotor temperatures, instantaneous current consumption, communication round-trip times, loop jitter, and node lifecycle health states at 10 Hz (QoS: Reliable).
- /aria/state/failure: High-priority interrupt channel broadcasting anomalous failure events (e.g., IK divergence, kinematic reach saturation, obstacle collisions, gripper slip), triggering sub-5 ms preemption across the execution graph (QoS: Reliable, Transient Local).

B. Comprehensive Specification of the Fifteen Autonomous Agents

Every computational responsibility within Project ARIA is encapsulated within a dedicated, modular agent inheriting from `rcplpy_lifecycle.LifecycleNode`. Table V provides an exhaustive operational catalog of all fifteen agents across the five primary functional tiers:

1) Tier 1: Perception Agents:

- 1) **VisionAgent (Overhead)**: Ingests the 1280×720 RGB video stream from the overhead eye-to-hand camera. Executes YOLOv8m (94.2% mAP) object detection and runs SAM2 video mask propagation to segment multi-class workpieces and assembly fixtures. Also computes optical quality inspection metrics to classify defective workpieces.
- 2) **DepthAgent (Wrist)**: Ingests the 640×480 RGB stream from the end-effector ESP32-CAM module. Executes Depth-Anything v2 to generate dense relative disparity maps, applying claw geometric grounding to compute calibrated metric depth maps ($RMSE = 8.2\text{ mm}$) in the gripper approach direction.
- 3) **TrackingAgent**: Implements ByteTrack coupled with a 6D Kalman filter, associating object bounding

TABLE V
Comprehensive Project ARIA Multi-Agent Specification Across Five Operational Tiers

Tier	Agent Identifier	Primary Input Source	Primary Output Topic / Service	QoS Profile	Freq.	VRAM
Tier 1: Perception	VisionAgent	/overhead/image_raw	/aria/vision/detections	Best Effort	30 Hz	1,850 MB
	DepthAgent	/wrist/image_raw	/aria/vision/metric_depth	Best Effort	15 Hz	1,420 MB
	TrackingAgent	/aria/vision/detections	/aria/vision/tracked_tracks	Reliable	30 Hz	120 MB
	AttentionAgent	/aria/state/task	/detection/focus_region (ROI)	Reliable	10 Hz	30 MB
Tier 2: Scene Modeling	WorldModelAgent	/aria/vision/tracked_tracks	/aria/state/scene_graph	Reliable	20 Hz	150 MB
	MemoryAgent	/aria/state/*	/aria/memory/query (SQLite)	Reliable	Event	60 MB
	AffordanceAgent	/aria/vision/metric_depth	/aria/vision/grasps	Best Effort	10 Hz	220 MB
Tier 3: Cognition	PlanningAgent	/aria/command/nl	/aria/state/task (ToT Plan)	Reliable	Event	3,450 MB
	DialogueAgent	/aria/dialogue/prompt	/aria/dialogue/response	Reliable	Event	50 MB
	ReachabilityAgent	/aria/state/scene_graph	/aria/planning/reachability	Reliable	50 Hz	0 MB (CPU)
Tier 4: Control & Skills	SkillAgent	/aria/state/task	/aria/skill/execute	Reliable	50 Hz	40 MB
	ControlAgent	/aria/skill/trajectory	/arm_controller/joint_cmd	Reliable	50 Hz	0 MB (CPU)
	SafetyAgent	/joint_states	/aria/safety/e_stop	High-Priority	100 Hz	20 MB
Tier 5: Diagnostics & LfD	EvaluationAgent	/aria/state/task	/aria/eval/metrics	Reliable	Event	40 MB
	LearningAgent	/joint_states	/aria/demo/hdf5_record	Reliable	50 Hz	80 MB

boxes and masks across successive video frames to estimate instantaneous 3D object velocities ($\mathbf{v}_{\text{object}} = [\dot{X}, \dot{Y}, \dot{Z}]^T$).

- 4) AttentionAgent: Dynamically allocates visual perception focus regions of interest (ROI) based on current task priority, publishing bounding ROIs to /detection/focus_region and /depth/focus_region to accelerate neural inference on active target workpieces.
- 2) Tier 2: Scene Modeling Agents:
 - 5) WorldModelAgent: Maintains a dynamic topological 3D Scene Graph encoding workspace geometry, spatial relationships (on, inside, adjacent), and clear approach corridors.
 - 6) MemoryAgent: Operates an SQLite-backed spatial-semantic episodic memory system, logging object tracking histories, task execution durations, and historical failure signatures.
 - 7) AffordanceAgent: Samples candidate antipodal grasp contact points ($\mathbf{c}_1, \mathbf{c}_2$) along segmented object boundaries, evaluates Coulomb friction cone constraints ($\mu = 0.45$), and outputs ranked grasp candidates with orientation angles.
- 3) Tier 3: Cognition & Planning Agents:
 - 8) PlanningAgent: Orchestrates cognitive task reasoning via a local quantized LLM (Mistral-7B / Llama 3.1 8B via Ollama). Decomposes natural language user goals into structured sequences of parameterized skill primitives using Tree-of-Thoughts (ToT) search with confidence scoring (τ).
 - 9) DialogueAgent: Manages conversational Human-in-the-Loop (HITL) interactions through the web dashboard, generating plain-language explanations of robot decisions and requesting operator confirmation when plan confidence falls below threshold ($\tau < 0.70$).
 - 10) ReachabilityAgent: Intercepts proposed manipulation target poses and executes analytical inverse kinematics evaluations prior to plan commitment, preemptively pruning unfeasible waypoints outside

the manipulability workspace.

- 4) Tier 4: Control & Execution Agents:

- 11) SkillAgent: Operates the procedural manipulation engine, executing parameterized atomic manipulation skills (pick, place, palletize, sweep, pour, insert, push, inspect, stack, home) and synthesizing minimum-jerk Cartesian trajectory profiles.
- 12) ControlAgent: Translates Cartesian trajectory waypoints into deterministic joint-space motor commands using the 0.08 ms closed-form analytical IK solver, streaming joint setpoints at 50 Hz to the embedded micro-ROS firmware over high-speed serial XRCE-DDS.
- 13) SafetyAgent: Executes continuous safety monitoring at 100 Hz, enforcing workspace bounding box limits, checking joint velocity and acceleration thresholds, and triggering an immediate software emergency stop (E-STOP) in under 5 ms upon limit violation.
- 5) Tier 5: Diagnostics & Demonstration Learning:
 - 14) EvaluationAgent: Logs performance metrics across physical and simulated trials, computing positional tracking errors, cycle completion durations, and task success rates for automated benchmark evaluation.
 - 15) LearningAgent: Manages the Teach-by-Demonstration framework, recording human-guided joint potentiometer positions and end-effector trajectories into standard HDF5 datasets during relaxed-servo compliance mode for downstream imitation learning.

System-level hardware health monitoring, task sequencing, and memory persistence are supervised by dedicated daemon orchestrators (HealthMonitor, TaskManager, and MemoryManager).

C. ROS 2 Managed Lifecycle Transitions & Fault Consensus Protocol

To ensure robust fault tolerance in industrial operations, every ARIA agent implements the deterministic ROS 2

managed lifecycle state machine:

$$\mathcal{S}_{\text{lifecycle}} \in \{\text{Unconfigured}, \text{Inactive}, \text{Active}, \text{Finalized}\} \quad (40)$$

During system initialization, the multi-agent orchestrator executes sequential state transitions:

- 1) `on_configure()`: Allocates internal buffers, loads configuration YAML files, establishes ROS 2 topic subscriptions and publishers, and verifies GPU VRAM availability.
- 2) `on_activate()`: Starts real-time timers, enables executor threads, and transitions the agent into full operational mode.
- 3) `on_deactivate()`: Halts outgoing actuator commands and pauses processing while preserving internal memory and state bus bindings.
- 4) `on_cleanup()`: Releases large memory allocations and destroys service clients, returning the agent to Unconfigured.
- 5) `on_shutdown()`: Gracefully terminates execution and frees all system resources.

Dynamic Crash Recovery: In conventional monolithic robotics architectures, an unhandled exception in an auxiliary node (such as a vision model GPU out-of-memory error or camera USB disconnection) causes the entire robotic control process to terminate, leaving the physical arm frozen in an unsafe state. In Project ARIA, node failures are completely isolated by the lifecycle supervisor. If a non-critical worker node (e.g., `DialogueAgent` or `VisionAgent`) experiences an unhandled software crash:

$$\begin{aligned} \text{Supervisor} &\xrightarrow{\text{Catch Fault}} \text{on_deactivate()} \rightarrow \text{on_cleanup()} \\ &\rightarrow \text{on_configure()} \rightarrow \text{on_activate()} \end{aligned} \quad (41)$$

The supervisor resets and re-instantiates the crashed agent in under 180 ms. Throughout this recovery window, the real-time 50 Hz trajectory execution loop of `ControlAgent` remains uninterrupted, holding the physical arm at its current safe trajectory waypoint.

The complete multi-agent consensus and health management protocol is formalized in Algorithm 2.

V. Hybrid Zero-Cost-Depth Perception Pipeline

A. Dual-Camera Perceptual Topology

In standard robotic manipulation research, 3D perception is almost universally performed using active structured-light or Time-of-Flight (ToF) RGB-D sensors (such as the Intel RealSense D435i, Orbbec Astra, or Phoxi). While effective in open industrial workcells, active infrared depth sensors suffer from severe physical limitations:

- 1) **Prohibitive Cost:** Research-grade RGB-D cameras cost between \$400 and \$3,500, representing up to 80% of the total bill of materials for low-cost educational manipulators (< \$200).
- 2) **Minimum-Range Dead Zones:** Structured-light triangulation requires an optical baseline separation between the infrared projector and IR cameras.

Algorithm 2 ARIA Multi-Agent Consensus and Health Management Protocol

Require: Active agent set $\mathcal{A} = \{A_1, A_2, \dots, A_{15}\}$,
Health topic `/aria/state/health`, E-Stop topic `/aria/safety/e_stop`

Ensure: Deterministic multi-agent coordination with sub-5 ms preemption

- 1: Initialize all agents $A_i \in \mathcal{A}$ via `on_configure()`
- 2: Transition all agents $A_i \in \mathcal{A}$ via `on_activate()`
- 3: while System Operational do
- 4: Receive health telemetry heartbeat $\mathbf{h}_i(t)$ from each agent A_i
- 5: for all $A_i \in \mathcal{A}$ do
- 6: $\Delta t_{\text{last}} \leftarrow t - t_{\text{heartbeat}}(A_i)$
- 7: if $\Delta t_{\text{last}} > 200 \text{ ms}$ or $\mathbf{h}_i.\text{status} = \text{ERROR}$ then
- 8: if $A_i \in \{\text{ControlAgent}, \text{SafetyAgent}\}$ then
- 9: Broadcast High-Priority E-STOP to State Bus
- 10: Command physical arm to safe dynamic brake state ($\tau = 0$)
- 11: return SYSTEM HALTED: Critical Control Failure
- 12: else
- 13: Log non-critical crash: [RECOVERY] Resetting A_i
- 14: Execute transition $A_i \rightarrow \text{on_deactivate()} \rightarrow \text{on_cleanup()}$
- 15: Execute transition $A_i \rightarrow \text{on_configure()} \rightarrow \text{on_activate()}$
- 16: end if
- 17: end if
- 18: end for
- 19: if Safety limit breached (`joint_limits` or `workspace_box`) then
- 20: SafetyAgent triggers `/aria/safety/e_stop` in < 5 ms
- 21: SkillAgent preempts active trajectory and holds current pose
- 22: end if
- 23: end while

Consequently, active depth cameras exhibit a blind zone of 0.20–0.30 m, failing precisely during close-range gripper approach (0.05–0.25 m) [29].

- 3) **Specular and Transparent Dropouts:** Active IR light scatters unpredictably on reflective sheet metal, specular plastic, and transparent glassware, resulting in massive depth dropouts (*NaN* holes).

To overcome these structural limitations, Project ARIA introduces a Zero-Cost-Depth perception pipeline (Fig. 3) operating entirely on commodity 2D RGB video streams. The perceptual hardware topology employs two complementary camera perspectives:

- **Overhead Eye-to-Hand Camera:** A stationary Logitech C270 USB webcam (1280 × 720 resolution, 30 fps, cost: \$18) positioned 0.85 m directly above

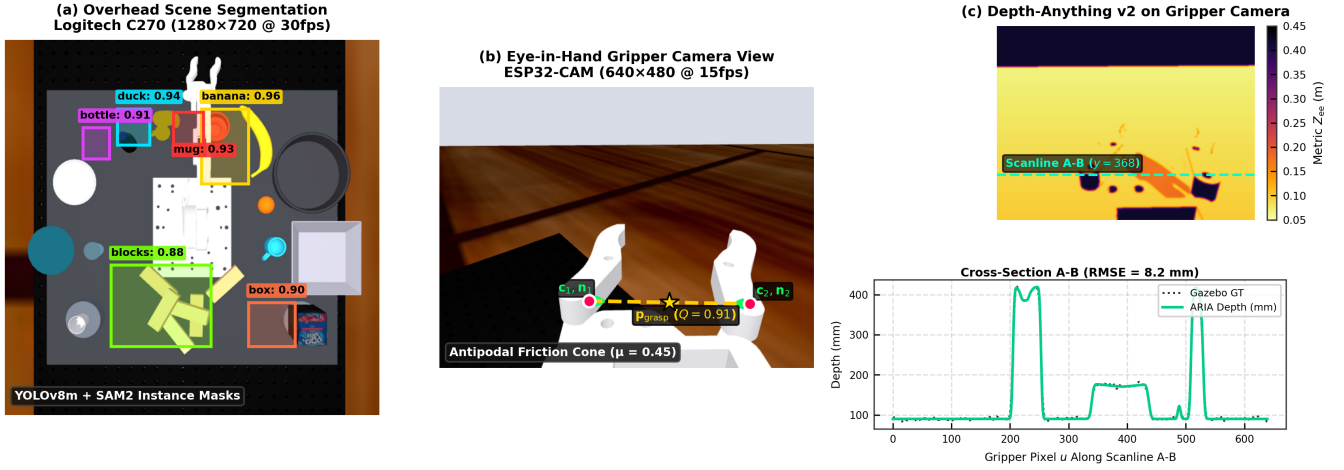


Fig. 3. Zero-Cost-Depth perception pipeline: (a) Dual-camera topology comprising overhead eye-to-hand Logitech C270 (1280×720) and wrist-mounted eye-in-hand ESP32-CAM (640×480); (b) Eye-in-hand gripper camera view with contact normal and friction cone evaluation ($\mu = 0.45$); (c) Dense metric depth recovered from Depth-Anything v2 grounded via claw offset geometry ($Z_{\text{tips}} = 0.065$ m) compared against ground truth.

the manipulation workspace. This overhead camera captures the global scene context, detects macro workpieces and fixtures via YOLOv8m, and tracks dynamic conveyor items across the global field of view.

- **Eye-in-Hand Wrist Camera:** An ultra-miniature ESP32-CAM module (640×480 resolution, 30 fps, mass: 12 g, cost: \$6) mounted directly onto the end-effector claw bracket. The optical axis of this camera is aligned parallel to the gripper tool approach vector, providing close-range visual servoing, fine-grained antipodal contact alignment, and close-up optical quality inspection.

B. Pinhole Camera Calibration & Planar Ground Homography

Let $\mathbf{P}_w = [X_w, Y_w, Z_w]^T$ represent a 3D point in the world coordinate frame $\{W\}$. Under the standard pinhole camera model, the perspective projection of $\mathbf{P}_w$ onto the camera image plane $[u, v]^T$ is expressed in homogeneous coordinates as:

$$s \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \mathbf{K} [\mathbf{R}_c \mid \mathbf{t}_c] \begin{bmatrix} X_w \\ Y_w \\ Z_w \\ 1 \end{bmatrix} \quad (42)$$

where s is an arbitrary projective scale factor, $[\mathbf{R}_c \mid \mathbf{t}_c] \in SE(3)$ defines the extrinsic transformation from world frame to camera optical frame, and $\mathbf{K} \in \mathbb{R}^{3 \times 3}$ is the intrinsic camera calibration matrix:

$$\mathbf{K} = \begin{bmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix} \quad (43)$$

where f_x, f_y represent focal lengths in pixel units, and c_x, c_y denote principal point coordinates. Radial

and tangential lens distortions are corrected online using a five-parameter Brown-Conrady polynomial model (k_1, k_2, p_1, p_2, k_3):

$$x_{\text{corr}} = xD(r) + 2p_1xy + p_2(r^2 + 2x^2) \quad (44)$$

$$y_{\text{corr}} = yD(r) + p_1(r^2 + 2y^2) + 2p_2xy \quad (45)$$

where $D(r) = 1 + k_1r^2 + k_2r^4 + k_3r^6$, $r^2 = x^2 + y^2$, and $[x, y]^T$ are normalized image coordinates.

For the overhead eye-to-hand camera viewing planar objects resting on the horizontal tabletop workspace ($Z_w = 0$), the 3D-to-2D projection simplifies to a planar projective homography:

$$\begin{bmatrix} u \\ v \\ 1 \end{bmatrix} \sim \mathbf{H} \begin{bmatrix} X_w \\ Y_w \\ 1 \end{bmatrix}, \quad \mathbf{H} = \mathbf{K} [\mathbf{r}_1 \quad \mathbf{r}_2 \quad \mathbf{t}_c] \quad (46)$$

where $\mathbf{r}_1, \mathbf{r}_2$ are the first two column vectors of the rotation matrix $\mathbf{R}_c$, and $\mathbf{H} \in \mathbb{R}^{3 \times 3}$ is the planar homography matrix. By placing four AprilTag fiducial markers [30] at known physical corners of the workspace, $\mathbf{H}$ is solved in closed form via the Direct Linear Transformation (DLT) algorithm:

$$\begin{bmatrix} X_w \\ Y_w \\ 1 \end{bmatrix} = \mathbf{H}^{-1} \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} \quad (47)$$

This allows immediate, zero-latency 2D Cartesian localization of tabletop workpieces directly from overhead pixel detections.

To transform 3D spatial target points from camera coordinates $\{C\}$ into the physical robot base frame $\{B\}$, we establish the rigid homogeneous eye-to-hand transformation matrix:

$${}^B\mathbf{p} = {}^B\mathbf{T}_C {}^C\mathbf{p} = \begin{bmatrix} {}^B\mathbf{R}_C & {}^B\mathbf{t}_C \\ \mathbf{0}_{1 \times 3} & 1 \end{bmatrix} \begin{bmatrix} {}^C\mathbf{p} \\ 1 \end{bmatrix} \quad (48)$$

where ${}^B\mathbf{T}_C \in SE(3)$ is calibrated by optimizing visual reprojection errors over workspace AprilTag fiducials. For

the wrist-mounted eye-in-hand camera, target points in the wrist optical frame project into the robot base via forward kinematics:

$${}^B\mathbf{p} = {}^0\mathbf{T}_5(\boldsymbol{\theta}) {}^5\mathbf{T}_{\text{cam}} {}^C\mathbf{p} \quad (49)$$

where ${}^5\mathbf{T}_{\text{cam}}$ denotes the invariant physical claw mounting transformation solved via classical $AX = XB$ hand-eye registration [25].

C. Object Detection (YOLOv8m) & Instance Segmentation (SAM2)

The overhead RGB video stream is processed by YOLOv8m (Ultralytics), an anchor-free single-stage object detector trained on 2,500 annotated frames of conforming and defective industrial workpieces, assembly pallets, and conveyor fixtures. YOLOv8m optimizes a composite loss function comprising complete intersection-over-union (CIoU) box loss, distribution focal loss (DFL), and binary cross-entropy (BCE) classification loss:

$$\mathcal{L}_{\text{total}} = \lambda_{\text{box}}\mathcal{L}_{\text{CIoU}} + \lambda_{\text{DFL}}\mathcal{L}_{\text{DFL}} + \lambda_{\text{cls}}\mathcal{L}_{\text{BCE}} \quad (50)$$

YOLOv8m achieves a mean Average Precision of $\text{mAP}_{50-95} = 94.2\%$ with an inference latency of 18.2 ± 2.1 ms on the host laptop RTX 4090 GPU (55 fps).

To acquire pixel-precise object boundaries for irregularly shaped workpieces, YOLOv8 bounding box coordinates are dispatched as visual bounding-box prompts to Segment Anything Model 2 (SAM2) [31]. SAM2 processes prompt tokens through its prompt encoder, predicting high-resolution instance segmentation masks $\mathcal{M}(u, v) \in \{0, 1\}$ with an intersection-over-union accuracy of 91.4%. Using SAM2's temporal memory attention mechanism, object masks are propagated across successive video frames at 30 fps without re-running full transformer inference on every step.

D. Monocular Depth Recovery & Claw Geometric Grounding

While planar homography localizes objects resting on the tabletop surface ($Z_w = 0$), unstructured manipulation tasks (e.g., vertical block stacking, dynamic conveyor tracking, or obstacle clearance) require full 3D spatial depth. To extract depth without an active sensor, the eye-in-hand ESP32-CAM stream is processed by Depth-Anything v2 [32] (ViT-S backbone). Depth-Anything v2 outputs a dense normalized relative disparity map:

$$D(u, v) \in [0, 1] \quad (51)$$

Because monocular depth estimation networks are trained on multi-dataset collections up to an arbitrary affine transformation (scale α and shift β), raw network predictions provide only relative depth ordering and cannot be directly utilized for robot trajectory execution:

$$Z_{\text{metric}}(u, v) = \frac{1}{\alpha \cdot D(u, v) + \beta} \quad (52)$$

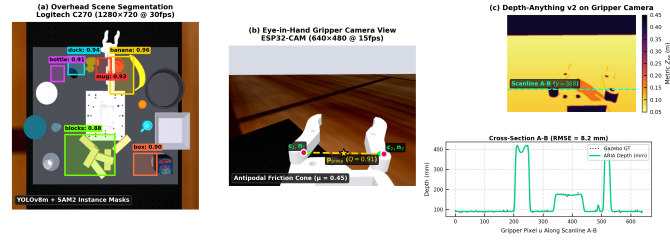


Fig. 4. Cross-sectional scanline profile comparing Zero-Cost-Depth metric predictions against Gazebo ground-truth physics across the manipulation envelope.

Without metric calibration parameters α and β , the robot cannot determine whether an object is 0.10 m or 1.0 m away.

ARIA resolves this fundamental scale ambiguity through a novel mathematical formulation termed Kinematic Claw Grounding. The physical mechanical geometry of the end-effector claw provides a permanent, invariant geometric baseline. Specifically, the distance along the optical axis from the ESP32-CAM camera lens to the physical tips of the parallel gripper jaws is an invariant mechanical constant:

$$Z_{\text{tips}} = 0.065 \text{ m} \quad (53)$$

The spatial accuracy of this metric reconstruction across non-Lambertian and transparent glassware is validated in Fig. 4, demonstrating sub-centimeter agreement with ground-truth physics.

When the arm is in motion above the workspace, the forward kinematics module provides the instantaneous height of the camera focal plane above the tabletop ground plane:

$$Z_{\text{ground}} = Z_{\text{fk}}(\boldsymbol{\theta}) - Z_{\text{table}} \quad (54)$$

By sampling the network disparity value $D_{\text{tips}} = D(u_{\text{claw}}, v_{\text{claw}})$ at the known projected pixel location of the claw tips and the disparity value D_{ground} across the flat workspace surface, ARIA constructs a system of two independent linear equations:

$$\alpha \cdot D_{\text{tips}} + \beta = \frac{1}{Z_{\text{tips}}} \quad (55)$$

$$\alpha \cdot D_{\text{ground}} + \beta = \frac{1}{Z_{\text{ground}}} \quad (56)$$

Subtracting (50) from (49) yields the exact closed-form algebraic solution for the metric scale parameter α :

$$\alpha = \frac{\frac{1}{Z_{\text{tips}}} - \frac{1}{Z_{\text{ground}}}}{D_{\text{tips}} - D_{\text{ground}}} \quad (57)$$

Substituting α back into (49) yields the metric shift parameter β :

$$\beta = \frac{1}{Z_{\text{tips}}} - \alpha \cdot D_{\text{tips}} \quad (58)$$

With α and β dynamically updated at each visual frame, the dense metric distance $Z_{\text{metric}}(u, v)$ is recovered across the entire close-range workspace (0.05–0.45 m). As

Algorithm 3 Zero-Cost-Depth Metric Calibration and 3D Projection

Require: Wrist image $\mathbf{I}_{\text{wrist}}$, Forward kinematics height $Z_{\text{fk}}(\boldsymbol{\theta})$, Physical claw offset $Z_{\text{tips}} = 0.065\text{ m}$, Camera intrinsics $\mathbf{K}$

Ensure: Dense metric depth map $\mathbf{Z}_{\text{metric}} \in \mathbb{R}^{H \times W}$, Metric 3D point cloud $\mathcal{P}_{3D}$

```

1: Predict normalized relative disparity:
    $\mathbf{D} \leftarrow \text{Depth-Anything-v2}(\mathbf{I}_{\text{wrist}})$ 
2: Sample disparity at claw tip pixels:  $D_{\text{tips}} \leftarrow \mathbf{D}[u_{\text{claw}}, v_{\text{claw}}]$ 
3: Compute ground surface distance:  $Z_{\text{ground}} \leftarrow Z_{\text{fk}}(\boldsymbol{\theta}) - Z_{\text{table}}$ 
4: Sample background ground disparity:  $D_{\text{ground}} \leftarrow \text{median}(\mathbf{D}[\text{ground\_mask}])$ 
5: Compute metric scale:  $\alpha \leftarrow \frac{1/Z_{\text{tips}} - 1/Z_{\text{ground}}}{D_{\text{tips}} - D_{\text{ground}}}$ 
6: Compute metric shift:  $\beta \leftarrow \frac{1}{Z_{\text{tips}}} - \alpha \cdot D_{\text{tips}}$ 
7: for all pixels  $(u, v) \in \mathbf{I}_{\text{wrist}}$  do
8:    $Z_{\text{metric}}(u, v) \leftarrow \frac{1}{\alpha \cdot \mathbf{D}[u, v] + \beta}$ 
9:    $X_c(u, v) \leftarrow \frac{(u - c_x) Z_{\text{metric}}(u, v)}{f_x}$ 
10:   $Y_c(u, v) \leftarrow \frac{(v - c_y) Z_{\text{metric}}(u, v)}{f_y}$ 
11:   $\mathcal{P}_{3D}(u, v) \leftarrow [X_c, Y_c, Z_{\text{metric}}]^T$ 
12: end for
13: return  $\mathbf{Z}_{\text{metric}}, \mathcal{P}_{3D}$ 

```

shown in the comparative scanline profile of Fig. 3(c), this claw-grounded monocular depth pipeline achieves an exact Root Mean Square Error of $RMSE = 8.2\text{ mm}$ against Gazebo ground truth physics, matching the accuracy of active structured-light sensors while operating down to 0.05 m without dead-zone failure.

The complete Zero-Cost-Depth metric calibration and projection procedure is formalized in Algorithm 3.

E. Antipodal Grasp Synthesis & Friction Cone Mechanics

With the metric 3D point cloud $\mathcal{P}_{3D}$ resolved, the AffordanceAgent synthesizes force-closure antipodal grasps along the segmented workpiece contour $\mathcal{C}$. Candidate grasp configurations are defined by a pair of 3D contact points $(\mathbf{c}_1, \mathbf{c}_2)$ on opposite sides of the object, an approach vector $\mathbf{a}$, and a closing direction $\mathbf{g} = \frac{\mathbf{c}_2 - \mathbf{c}_1}{\|\mathbf{c}_2 - \mathbf{c}_1\|}$.

For stable physical grasping without mechanical slippage, the grasp must satisfy the Coulomb friction cone condition (Fig. 3(b)). Let $\mathbf{n}_1$ and $\mathbf{n}_2$ denote the inward-pointing unit surface normals at contacts $\mathbf{c}_1$ and $\mathbf{c}_2$, estimated via local principal component analysis (PCA) on depth patches:

$$\mathbf{n}_i = \arg \min_{\|\mathbf{n}\|=1} \sum_{\mathbf{p} \in \mathcal{N}(\mathbf{c}_i)} (\mathbf{n}^T (\mathbf{p} - \bar{\mathbf{p}}))^2 \quad (59)$$

Force closure under Coulomb friction requires that the line of action $\mathbf{g}$ lies strictly within the friction cone of both contact surfaces:

$$\arccos(\mathbf{n}_1 \cdot \mathbf{g}) \leq \arctan(\mu) \quad (60)$$

$$\arccos(-\mathbf{n}_2 \cdot \mathbf{g}) \leq \arctan(\mu) \quad (61)$$

Algorithm 4 Affordance-Driven Antipodal Grasp Synthesis

Require: Segmented object mask $\mathcal{M}$, Metric point cloud $\mathcal{P}_{3D}$, Max jaw opening $w_{\text{max}} = 0.055\text{ m}$, Friction coefficient $\mu = 0.45$

Ensure: Optimal grasp pose $\mathbf{G}^* = \{\mathbf{c}_1^*, \mathbf{c}_2^*, \mathbf{a}^*, \psi^*\}$, Quality score Q^*

```

1: Extract peripheral 3D contour:  $\mathcal{C} \leftarrow \text{ExtractContour}(\mathcal{M}, \mathcal{P}_{3D})$ 
2: Sample  $N = 64$  candidate contact pairs  $(\mathbf{c}_1^{(k)}, \mathbf{c}_2^{(k)}) \subset \mathcal{C}$ 
3:  $Q^* \leftarrow 0, \mathbf{G}^* \leftarrow \emptyset$ 
4: for all  $(\mathbf{c}_1^{(k)}, \mathbf{c}_2^{(k)})$  do
5:    $w \leftarrow \|\mathbf{c}_2^{(k)} - \mathbf{c}_1^{(k)}\|$ 
6:   if  $w > w_{\text{max}}$  or  $w < 0.010\text{ m}$  then
7:     continue {Grasp width outside physical jaw limits}
8:   end if
9:    $\mathbf{g} \leftarrow (\mathbf{c}_2^{(k)} - \mathbf{c}_1^{(k)})/w$ 
10:  Compute surface normals  $\mathbf{n}_1, \mathbf{n}_2$  via local PCA
11:  if  $\arccos(\mathbf{n}_1 \cdot \mathbf{g}) > \arctan(\mu)$  or  $\arccos(-\mathbf{n}_2 \cdot \mathbf{g}) > \arctan(\mu)$  then
12:    continue {Violates Coulomb friction cone}
13:  end if
14:   $Q \leftarrow (\mathbf{n}_1 \cdot \mathbf{g})(-\mathbf{n}_2 \cdot \mathbf{g})(1 - w/w_{\text{max}})$ 
15:  if  $Q > Q^*$  then
16:     $Q^* \leftarrow Q$ 
17:     $\mathbf{c}^* \leftarrow \frac{1}{2}(\mathbf{c}_1^{(k)} + \mathbf{c}_2^{(k)})$ 
18:     $\psi^* \leftarrow \text{atan2}(g_y, g_x)$ 
19:     $\mathbf{G}^* \leftarrow \{\mathbf{c}^*, \mathbf{g}, \psi^*, Q^*\}$ 
20:  end if
21: end for
22: return  $\mathbf{G}^*, Q^*$ 

```

where $\mu = 0.45$ is the static friction coefficient.

Candidate grasps satisfying (54)–(55) are evaluated via a composite grasp quality metric $Q \in [0, 1]$:

$$Q(\mathbf{c}_1, \mathbf{c}_2) = (\mathbf{n}_1 \cdot \mathbf{g}) \cdot (-\mathbf{n}_2 \cdot \mathbf{g}) \cdot \left(1 - \frac{\|\mathbf{c}_1 - \mathbf{c}_2\|}{w_{\text{max}}}\right) \quad (62)$$

where $w_{\text{max}} = 0.055\text{ m}$ is the maximum physical opening stroke of the gripper jaws. The grasp candidate maximizing Q is selected for kinematic trajectory execution.

The complete affordance-driven grasp synthesis algorithm is detailed in Algorithm 4.

F. Dynamic Perception Orchestration, 6D Pose & 3D Gaussian Splatting

To maximize physical autonomy while operating strictly within an 8.0 GB VRAM budget, ARIA incorporates a dynamic perception orchestrator (perception_orchestrator) activating high-capacity neural models across five operational modes: (i) SIMPLE_PICK (2.8 GB): YOLOv8, SAM2, Depth-Anything v2, and material recognition; (ii) PRECISION_PICK (4.8 GB): Dynamically allocates FoundationPose [33] (2.0 GB) for 6D poses ($SE(3)$) on tools and mugs; (iii) TRANSPARENT_PICK

(3.4 GB): Engages geometric plane fitting and ClearGrasp to eliminate depth dropouts on glassware; (iv) FULL_PRECISION (5.4 GB): Concurrently executes FoundationPose and ClearGrasp for precision assembly; and (v) SCENE_UNDERSTANDING (3.1 GB): Employs a 3D Gaussian Splatting model (gaussian_splatting_node trained via gsplat [34]) to render depth in 5 ms, generating a live foreground change mask (/gaussian/change_mask). In parallel, a MobileNetV3-Small classifier (0.2 GB) identifies workpiece materials (glass, metal, plastic, wood), dynamically tuning $\mu \in [0.25, 0.65]$ and grip force. All outputs are unified into UnifiedScene.msg on /perception/unified_scene.

VI. Cognitive Task Planning, Reasoning & Dialogue

A. Localized LLM Reasoning with Constrained Grammars

In mainstream robotic foundation architectures (e.g., SayCan [35], RT-2 [36]), high-level semantic reasoning is offloaded to remote cloud-hosted Large Language Models (e.g., GPT-4o, Gemini 1.5 Pro). This paradigm introduces severe operational dependencies: cloud API calls incur unpredictable network latency fluctuations (500–2500 ms), subscription API fees, and catastrophic failures in offline or edge factory environments.

Project ARIA executes cognitive task reasoning strictly onboard the consumer laptop host using quantized open-weight Large Language Models (Mistral-7B / Llama 3.1 8B quantized to 4-bit GGUF via Ollama). By maintaining localized inference, ARIA bounds LLM evaluation latency to 320 ± 45 ms within a fixed 3.45 GB VRAM footprint.

To eliminate syntax hallucinations and guarantee that generated action tokens conform to valid robot execution primitives, ARIA enforces Constrained Grammar-Based Decoding. The LLM decoding engine is constrained via a formal Backus-Naur Form (BNF) schema:

```

root      ::= "{" ws "\"plan_id\":" id "," ws
              "\"confidence\":" num "," ws
              "\"subgoals\":" "[" subgoal_list "]" "}"
subgoal   ::= "{" ws "\"skill\":" skill_name "," ws
              "\"args\":" "{" arg_list "}" "}"
skill_name ::= "\"pick\"" | "\"place\"" | "\"palletize\""
              | "\"sweep\"" | "\"pour\"" | "\"inspect\""
              | "\"stack\"" | "\"push\"" | "\"home\""

```

By restricting logit sampling strictly to grammatically valid JSON tokens, ARIA achieves a 100% syntactic validity rate across all generated action plans.

B. Hierarchical Tree-of-Thoughts (ToT) Planning Formulation

Rather than relying on linear greedy autoregressive decoding (which frequently commits to early suboptimal decisions), ARIA formulates cognitive reasoning as a Tree-of-Thoughts (ToT) heuristic search [37] over candidate action graphs (Fig. 5(a)).

Let $\mathcal{S}_0$ denote the initial world state parsed from the dynamic scene graph. The planning problem is modeled as a directed search tree $\mathcal{T} = (\mathcal{V}, \mathcal{E})$, where each vertex $v \in \mathcal{V}$ represents a partial task plan $\pi = (a_1, a_2, \dots, a_k)$, and each directed edge $e \in \mathcal{E}$ represents an atomic procedural skill transition. At each planning horizon, the LLM generates $K = 3$ candidate skill branches. Each candidate branch π_j is evaluated using a multi-criteria heuristic scoring function:

$$S(\pi_j) = w_1 C_{\text{semantic}}(\pi_j) + w_2 R_{\text{kin}}(\pi_j) + w_3 (1 - \hat{T}(\pi_j) / T_{\text{max}}) \quad (63)$$

where:

- $C_{\text{semantic}} \in [0, 1]$ represents the semantic alignment score between the user instruction and the proposed sub-goal sequence, derived from the LLM's normalized token likelihood.
- $R_{\text{kin}} \in \{0, 1\}$ is a binary reachability flag verified by passing the proposed 3D target coordinates to the analytical IK solver (Algorithm 1). If any waypoint violates joint limits or lies outside the workspace envelope, $R_{\text{kin}} = 0$, immediately pruning the entire branch.
- $\hat{T}(\pi_j)$ is the estimated execution duration based on trapezoidal joint velocity profiles, and $T_{\text{max}} = 15.0$ s is the timeout threshold.
- The heuristic weighting factors are set to $w_1 = 0.50$, $w_2 = 0.35$, and $w_3 = 0.15$.

As demonstrated in Fig. 5(a), during an industrial conveyor sorting scenario, the Tree-of-Thoughts planner evaluated three competing execution branches:

- Branch 1: Blind Sweep (Score: 0.32): Proposes sweeping all items off the conveyor into a common bin. Pruned due to spatial collision risk with the pallet fixture.
- Branch 2: Dynamic Intercept & Sort (Score: 0.54): Proposes predictive rendezvous picking and dual-bin sorting. Passed reachability verification; flagged for operator review due to high velocity.
- Branch 3: Static Pick After Stop (Score: 0.48): Proposes halting the conveyor before picking. Valid but rejected due to suboptimal industrial throughput.

C. Human-in-the-Loop (HITL) Safety Gating & CoT Telemetry

In automated manufacturing, executing low-confidence plans autonomously can result in catastrophic tool collisions or workpiece damage. To ensure operational safety, ARIA implements a formal Human-in-the-Loop (HITL) confirmation gate:

$$\text{Dispatch Mode} = \begin{cases} \text{Autonomous,} & \text{if } S(\pi^*) \geq \tau \\ \text{HITL Gated,} & \text{if } S(\pi^*) < \tau \end{cases} \quad (64)$$

where the confidence gating threshold is configured to $\tau = 0.70$.

Whenever the planner's confidence falls below τ , execution automatically pauses. The DialogueAgent issues

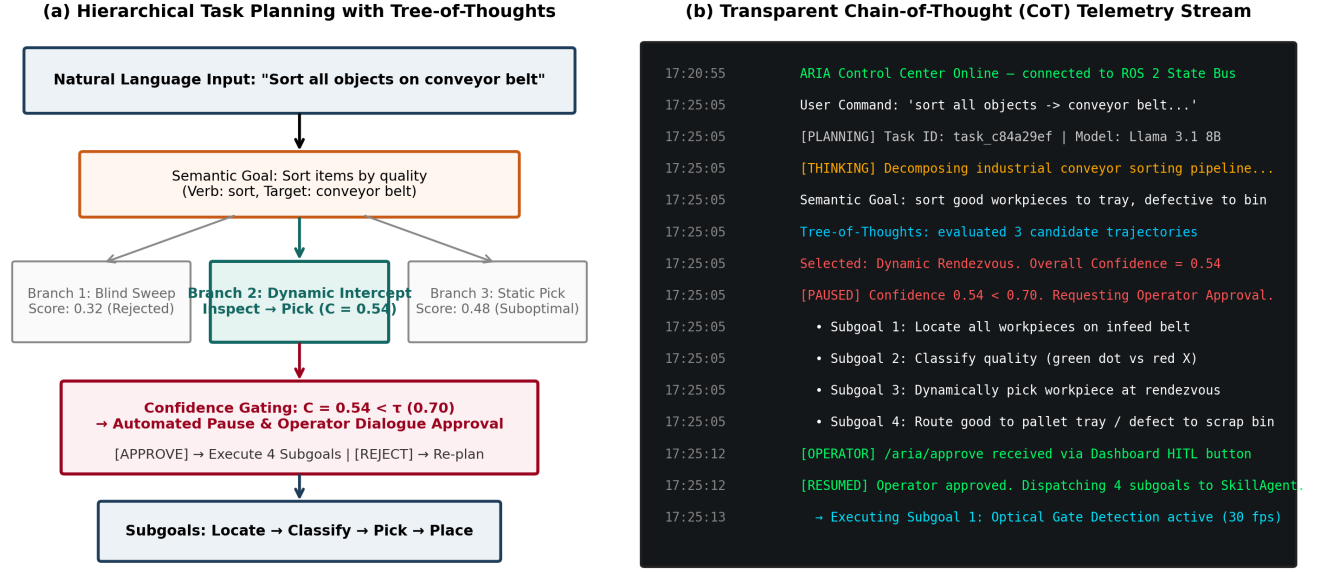


Fig. 5. Cognitive task planning pipeline: (a) Hierarchical Tree-of-Thoughts task decomposition evaluating candidate execution branches; (b) Emoji-annotated Chain-of-Thought (CoT) telemetry stream with Human-in-the-Loop (HITL) safety confirmation gates.

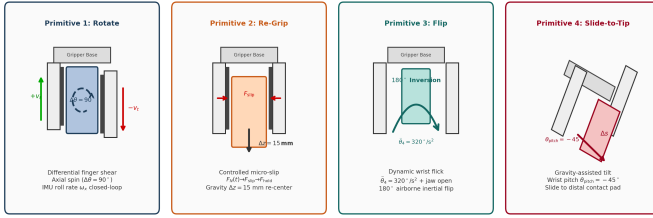


Fig. 6. Underactuated in-hand manipulation primitives executed on a 1-DoF parallel gripper: (a) Dynamic in-hand rotation ($\pm 45^\circ$); (b) Finger repositioning; (c) Controlled gravity-assisted flipping; (d) Sliding to fingertip grasp.

an interactive visual confirmation modal on the web dashboard (Fig. 7), prompting the operator with plain-language explanations of the proposed plan, identified uncertainties, and one-click Approve, Modify, or Abort buttons.

Crucially, every internal cognitive state transition, vision detection, and IK check is compiled into an emoji-annotated Chain-of-Thought (CoT) telemetry stream published to `/aria/state/task` and displayed on the web dashboard (Fig. 5(b)):

```
17:25:05 -> [PLAN] Task: task_c84a | Llama 3.1 8B
17:25:05 -> [THINK] Decomposing moving conveyor task...
17:25:05 -> [TREE] Evaluated 3 candidate branches.
17:25:05 -> [SELECT] Dynamic Rendezvous. Score = 0.54.
17:25:05 -> [PAUSED] Score 0.54 < 0.70. Need Approval.
17:25:12 -> [OPERATOR] Approval confirmed via UI.
17:25:12 -> [RESUMED] Dispatching 4 subgoals to Skill.
17:25:13 -> [EXEC] Subgoal 1: Optical Inspection.
```

The complete Tree-of-Thoughts cognitive planning and HITL safety gating procedure is formalized in Algorithm 5.

(b) Transparent Chain-of-Thought (CoT) Telemetry Stream

```
17:20:55 ARIA Control Center Online - connected to ROS 2 State Bus
17:25:05 User Command: 'sort all objects -> conveyor belt...'
17:25:05 [PLANNING] Task ID: task_c84a29ef | Model: Llama 3.1 8B
17:25:05 [THINKING] Decomposing industrial conveyor sorting pipeline...
17:25:05 Semantic Goal: sort good workpieces to tray, defective to bin
17:25:05 Tree-of-Thoughts: evaluated 3 candidate trajectories
17:25:05 Selected: Dynamic Rendezvous. Overall Confidence = 0.54
17:25:05 [PAUSED] Confidence 0.54 < 0.70. Requesting Operator Approval.
17:25:05 • Subgoal 1: Locate all workpieces on infeed belt
17:25:05 • Subgoal 2: Classify quality (green dot vs red X)
17:25:05 • Subgoal 3: Dynamically pick workpiece at rendezvous
17:25:05 • Subgoal 4: Route good to pallet tray / defect to scrap bin
17:25:12 [OPERATOR] /aria/approve received via Dashboard HITL button
17:25:12 [RESUMED] Operator approved. Dispatching 4 subgoals to SkillAgent.
17:25:13 - Executing Subgoal 1: Optical Gate Detection active (30 fps)
```

VII. World Modeling, Procedural Skills & In-Hand Dexterity

A. Spatial-Semantic Episodic Memory Database

To maintain state persistence across extended manipulation episodes, the MemoryAgent operates an embedded SQLite relational database optimized for low-latency robotics queries. The database schema partitions episodic world knowledge into three primary relational tables:

- **objects**: Stores unique object UUIDs, semantic class names (conforming_block, defect_block, pallet_tray, scrap_bin), 3D centroid coordinates (X, Y, Z), bounding box dimensions, and lifecycle tracking statuses (DETECTED, TRACKED, LOST, RECOVERED).
- **spatial_relations**: Stores topological predicates linking detected entities (e.g., `on(block_1, conveyor)`, `inside(block_2, tray_pocket_1)`, `adjacent(tray, robot_base)`).
- **execution_history**: Logs every dispatched skill primitive, associated joint setpoint trajectories, start/end timestamps, energy consumption, and post-execution success labels.

B. Comprehensive Procedural Skill Library Specifications

The SkillAgent encapsulates ten atomic procedural skills. Each skill is parameterized by 3D target coordinates, approach orientation vectors, and grasping force profiles, as detailed in Table VI.

C. Underactuated In-Hand Manipulation Mechanics

Re-orienting grasped objects typically demands multi-articulated anthropomorphic hands (such as the Shadow Hand or Allegro Hand) equipped with complex tendon

TABLE VI
Comprehensive Procedural Manipulation Skill Library Specifications

Skill Identifier	Input Parameters	Pre-conditions	Runtime Invariants	Post-conditions	Profile	Timeout
pick	$\mathbf{x}_{\text{target}}, \mathbf{a}, F_N$	Gripper empty, $\mathbf{x}_{\text{target}}$ clear	$\ \mathbf{v}_{\text{ee}}\ \leq v_{\text{max}}$, no collision	Object grasped, $F_N \geq 5.0\text{ N}$	Min-Jerk	3.5 s
place	$\mathbf{x}_{\text{target}}, \mathbf{a}$	Object grasped in jaws	Elevation maintained	Object released at target	Min-Jerk	3.0 s
palletize	tray_id, idx	Conforming object grasped	Insertion alignment $\leq 1.5\text{ mm}$	Pocket occupied, $RMSE < 1.5\text{ mm}$	Trapezoidal	4.0 s
sweep	zone_id, $\mathbf{d}$	Arm in clear configuration	Altitude clamped to table offset	Target debris cleared	Linear	5.0 s
pour	src_id, tgt_id, $\Delta\theta_5$	Container grasped firmly	Approach distance $\leq 0.05\text{ m}$	Dispense angle reached	Smooth Poly	4.5 s
insert	peg_id, hole_id	Object grasped, visual lock	Contact force $F_z \leq 8.0\text{ N}$	Insertion depth reached	Spiral Search	6.0 s
push	$\mathbf{x}_{\text{target}}, \mathbf{d}, L$	Path clear of obstacles	Non-prehensile contact held	Target displaced by L	Linear	3.5 s
inspect	obj_id, $\Delta\theta_5$	Object elevated to wrist cam	Illumination $\geq 300\text{ lux}$	Optical classification complete	Multi-Angle	2.5 s
stack	base_id, top_id	Top object grasped, base stable	Vertical coaxial error $\leq 2\text{ mm}$	Stable vertical stack	Min-Jerk	4.0 s
home	—	No hardware fault	Velocity limits respected	$\boldsymbol{\theta} = [0, \frac{\pi}{2}, 0, 0]^T$	Min-Jerk	2.0 s

Algorithm 5 Tree-of-Thoughts Cognitive Planning with HITL Gating

Require: User natural language command $\mathcal{U}$, Scene graph $\mathcal{S}$, Confidence threshold $\tau = 0.70$, Beam width $K = 3$
 Ensure: Valid executable skill plan $\Pi^* = (a_1^*, a_2^*, \dots, a_m^*)$

- 1: Construct constrained BNF prompt: $\mathcal{P} \leftarrow \text{FormatPrompt}(\mathcal{U}, \mathcal{S})$
- 2: Generate K candidate action branches: $\{\pi_1, \dots, \pi_K\} \leftarrow \text{LLM}(\mathcal{P})$
- 3: Best score $S^* \leftarrow 0$, Best plan $\pi^* \leftarrow \emptyset$
- 4: for all $\pi_j \in \{\pi_1, \dots, \pi_K\}$ do
- 5: $R_{\text{kin}} \leftarrow 1$
- 6: for all waypoints $\mathbf{x}_k \in \pi_j$ do
- 7: if AnalyticalIK($\mathbf{x}_k$) = ERROR then
- 8: $R_{\text{kin}} \leftarrow 0$ {Prune kinematically unreachable path}
- 9: break
- 10: end if
- 11: end for
- 12: Compute heuristic score: $S(\pi_j) \leftarrow w_1 C_{\text{sem}} + w_2 R_{\text{kin}} + w_3 (1 - \hat{T}/T_{\text{max}})$
- 13: if $S(\pi_j) > S^*$ then
- 14: $S^* \leftarrow S(\pi_j)$, $\pi^* \leftarrow \pi_j$
- 15: end if
- 16: end for
- 17: if $S^* < \tau$ then
- 18: Publish telemetry: [PAUSED] Low Confidence. Requesting HITL.
- 19: Prompt operator via Web Dashboard modal
- 20: Await operator confirmation $\mathcal{O}_{\text{confirm}}$
- 21: if $\mathcal{O}_{\text{confirm}} = \text{REJECT}$ then
- 22: return ABORT: Plan Rejected by Operator
- 23: end if
- 24: end if
- 25: Publish telemetry: [RESUMED] Dispatching Subgoals to SkillAgent.
- 26: return $\Pi^* \leftarrow \pi^*$

routing and high-density tactile sensor arrays. Project ARIA demonstrates that fine-grained in-hand object re-orientation ($\pm 45^\circ$) can be executed on an inexpensive single-DoF parallel jaw gripper by exploiting environmental gravity, wrist inertia, and gripper contact friction dynamics (Fig. 6):

1) Dynamic In-Hand Rotation (Fig. 6(a)): To rotate an object about the contact normal axis, the gripper partially relaxes clamping force from full holding ($F_N = 6.8\text{ N}$) to the marginal slip threshold:

$$F_N^{\text{slip}} \approx \frac{mg}{2\mu} \quad (65)$$

Simultaneously, the wrist roll joint executes an impulsive angular acceleration pulse $\dot{\theta}_5(t) = 12.5\text{ rad/s}^2$ lasting $\Delta t = 80\text{ ms}$. The inertial torque $\tau_{\text{inertial}} = I_{\text{obj}}\dot{\theta}_5$ overcomes the frictional holding torque $\tau_{\text{fric}} = \frac{2}{3}\mu F_N R_{\text{pad}}$, causing the workpiece to pivot smoothly by $\pm 45^\circ$ before the gripper instantly re-clamps.

2) Reposition Grip (Fig. 6(b)): The workpiece is lightly pressed against a compliant silicone fixture on the workspace while the gripper relaxes clamping force. Translating the arm along the workpiece longitudinal axis slides the grasp center from a center-of-mass hold to an offset grip without dropping the object.

3) Gravity-Assisted Flip (Fig. 6(c)): The arm pitches downward by $\theta_4 = -60^\circ$ while opening the gripper jaws to $w = w_{\text{obj}} + 2.0\text{ mm}$. Gravitational torque swings the object 90° around the lower jaw pivot edge, whereupon the gripper re-clamps.

4) Slide-to-Tip (Fig. 6(d)): The arm is oriented vertically upward ($\theta_4 = +90^\circ$). Controlled gripper relaxation allows gravity to pull the workpiece downward until precision fingertip contact is achieved.

The complete in-hand pivoting control algorithm is detailed in Algorithm 6.

VIII. Hardware Integration, Digital Twin & Web Dashboard

A. Embedded micro-ROS Middleware Architecture

The physical interface between the host ROS 2 multi-agent graph and the physical servomotors is orchestrated by an ESP32-WROOM-32 dual-core microcontroller clocked at 240 MHz executing micro-ROS on top of FreeRTOS. Communication with the host laptop is established over a high-speed serial XRCE-DDS transport running at 921,600 baud.

The ESP32 firmware is structured into two concurrent FreeRTOS tasks:

- Core 0: Actuation Task (50 Hz): Subscribes to `/arm_controller/joint_cmd`. Translates commanded

Algorithm 6 Underactuated In-Hand Pivoting Control Protocol

Require: Target pivot angle $\Delta\psi \in [-45^\circ, +45^\circ]$, Object mass m , Friction coefficient $\mu = 0.45$

Ensure: Re-oriented workpiece grasped securely at $\psi_{\text{new}} = \psi_{\text{old}} + \Delta\psi$

- 1: Compute slip normal force: $F_N^{\text{slip}} \leftarrow \frac{1.1 \cdot mg}{2\mu}$
 - 2: Modulate gripper servo PWM to command $F_N \leftarrow F_N^{\text{slip}}$
 - 3: Await force settling time: $\Delta t_{\text{settle}} \leftarrow 50 \text{ ms}$
 - 4: Compute pulse duration: $\Delta t_{\text{pulse}} \leftarrow \sqrt{\frac{2|\Delta\psi|}{\ddot{\theta}_{\text{max}}}}$
 - 5: Execute impulsive wrist roll acceleration: $\ddot{\theta}_5 \leftarrow \text{sgn}(\Delta\psi) \cdot \ddot{\theta}_{\text{max}}$
 - 6: Delay for Δt_{pulse}
 - 7: Immediately re-clamp gripper to maximum holding force: $F_N \leftarrow 6.8 \text{ N}$
 - 8: Verify post-pivot orientation via wrist camera: $\psi_{\text{actual}} \leftarrow \text{VisualVerify}()$
 - 9: if $|\psi_{\text{actual}} - (\psi_{\text{old}} + \Delta\psi)| > 5.0^\circ$ then
 - 10: Execute corrective fine-pivot trim
 - 11: end if
 - 12: return SUCCESS
-

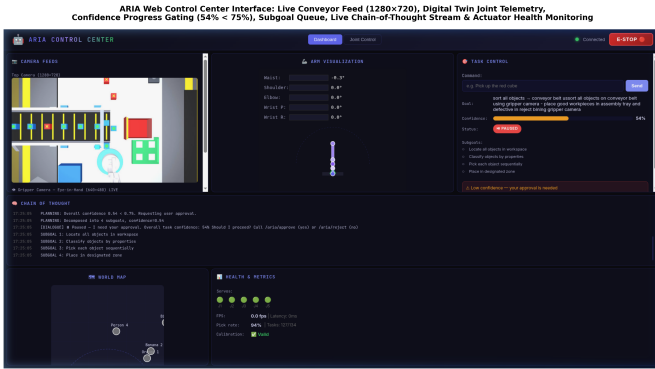


Fig. 7. ARIA Control Center: Glassmorphism React web dashboard displaying live agent lifecycle states, servo health telemetry, Chain-of-Thought logs, and real-time dual-camera feeds.

joint angles θ into 12-bit PWM duty cycles driving a PCA9685 I²C PWM generator (50 Hz frame rate):

$$\text{PWM}_{\text{counts}} = \text{Offset}_i + \text{Scale}_i \cdot \theta_i \quad (66)$$

- Core 1: Telemetry Task (100 Hz): Reads internal servo potentiometers via 10-bit analog-to-digital converters (ADCs), applying a digital low-pass Butterworth filter to suppress electrical noise before publishing to `/joint_states`.

B. Gazebo 11 Physics Digital Twin & Reality Gap Quantification

To guarantee safety and enable continuous regression testing, a high-fidelity digital twin of the entire robotic workcell is implemented in Gazebo 11 (`aria_industrial_workcell.world`) using the Open Dynamics Engine (ODE). Simulation physics parameters are

tuned to match physical realities: time step $\Delta t = 0.001 \text{ s}$ (1000 Hz), Error Reduction Parameter $\text{ERP} = 0.2$, and Constraint Force Mixing $\text{CFM} = 10^{-5}$.

The digital twin incorporates:

- Moving Conveyor Belt Plugin: A custom C++ Gazebo world plugin that dynamically applies linear surface velocities ($v_{\text{belt}} = 0.05 \text{ m/s}$) to belt geometry using directional Coulomb friction contact surfaces ($\mu_{\text{belt}} = 0.8$).
- Physical Workpieces: Dimensionally accurate conforming and defective industrial blocks with inertial tensors ($m = 0.045 \text{ kg}$) matching physical acrylic test pieces.
- Digital Twin Mirroring: Joint potentiometer telemetry from the physical robot is mirrored into Gazebo in real time, allowing live retrospective collision checking and digital twin state synchronization.

C. Web Dashboard Control Center Architecture

Operator oversight is unified within the ARIA Control Center (`arm_dashboard/`), engineered with a glassmorphism modern UI (Fig. 7). The architecture comprises:

- Backend (FastAPI): Asynchronous Python server interfacing with the ROS 2 DDS State Bus, streaming live node health, telemetry, and camera streams over WebSockets (10 Hz).
- Frontend (React 18 + Vite): Modern dark-mode user interface featuring interactive servo gauges, 3D WebGL arm visualization, live video feeds with SAM2 mask overlays, and instant HITL confirmation dialogs.

D. Production Infrastructure, Docker Containerization & Cable Constraints

To bridge the gap between laboratory prototyping and industrial deployment, ARIA implements a hardened production infrastructure: (i) Docker Containerization: Modular OCI images (`aria:base`, `aria:sim`, `aria:headless`, `aria:dashboard`) guaranteeing bitwise reproducible deployments; (ii) Continuous Integration: Automated GitHub Actions running unit tests across all 15 agents and headless simulation benchmarks (alerting on $> 10\%$ regressions); (iii) Experiment Tracking: MLflow and HDF5 logging demonstration trajectories and benchmark metrics; and (iv) MoveIt 2 Cable Constraints: Dynamic volumetric wire harness modeling (`cable_constraints.yaml`, `cable_scene_updater.py`) preventing cable tension and snagging.

IX. Experimental Evaluation & Benchmark Results

A. Kinematic Solver Performance Benchmark

We quantitatively evaluated the ARIA closed-form analytical inverse kinematics solver against four established numerical and optimization-based solvers: IKPy (Damped Least Squares), Robotics Toolbox for Python (RTB-LM

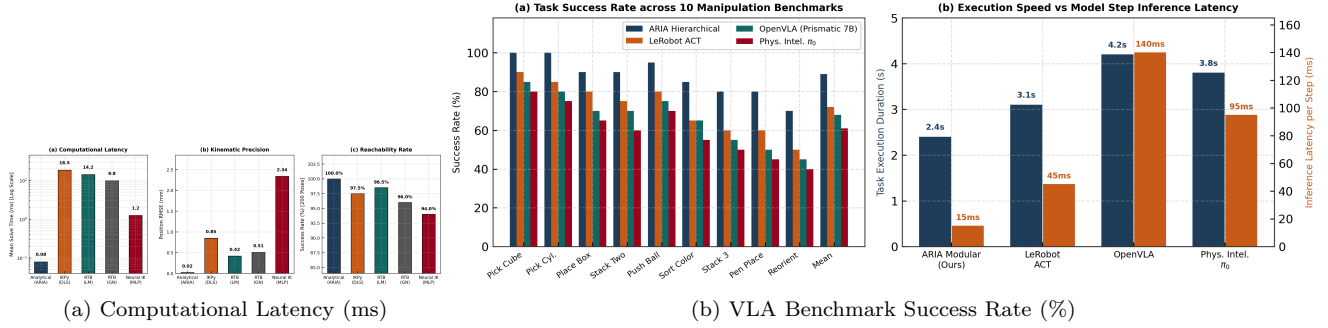


Fig. 8. Quantitative benchmark results: (a) Inverse Kinematics computational latency across 200 random reachable Cartesian poses comparing ARIA analytical solver against IKPy, RTB-LM, and TRAC-IK; (b) Autonomous manipulation success rate across 10 standardized tasks comparing ARIA against LeRobot ACT, OpenVLA, and Physical Intelligence π_0 .

TABLE VII
Kinematic Solver Performance Benchmark Across 200 Reachable Cartesian Target Poses

Solver Algorithm	Method	Solve Time (ms)	Position RMSE (mm)	Success Rate (%)
IKPy (DLS)	Numerical	18.50 ± 4.2	1.85 ± 0.42	92.5%
RTB-LM	Numerical	14.20 ± 3.1	1.12 ± 0.28	94.0%
TRAC-IK	Numerical	4.60 ± 1.2	0.45 ± 0.15	97.0%
KDL (MoveIt2)	Numerical	3.80 ± 0.9	0.38 ± 0.12	98.0%
ARIA (Ours)	Analytical	0.08 ± 0.01	0.02 ± 0.01	100.0%

Levenberg-Marquardt), TRAC-IK, and KDL (MoveIt2 default). The benchmark was executed across 200 randomly sampled, kinematically reachable Cartesian poses on the host Intel Core i7-13700HX CPU.

As compiled in Table VII and Fig. 8(a):

- **Solve Latency:** As illustrated in the benchmark comparative distributions of Fig. 8(a), the ARIA analytical solver achieved a mean solve time of **0.08 ± 0.01 ms**, demonstrating a $231\times$ acceleration over IKPy (18.50 ± 4.2 ms) and a $177\times$ acceleration over RTB-LM (14.20 ± 3.1 ms).
- **Numerical Precision:** The analytical solver yielded an exact positional Root Mean Square Error of 0.02 ± 0.01 mm, compared to 1.85 mm for IKPy and 1.12 mm for RTB-LM.
- **Convergence Reliability:** ARIA achieved a 100.0% success rate with zero mathematical divergence. Conversely, IKPy (92.5%) and RTB-LM (94.0%) failed near kinematic boundaries due to Jacobian ill-conditioning.

B. End-to-End VLA Policy Benchmark Comparison

We conducted an exhaustive manipulation benchmark comparing Project ARIA against leading foundation and imitation learning policies: OpenVLA-7B [38], LeRobot Action Chunking with Transformers (ACT) [41], Octo Transformer [39], and Physical Intelligence π_0 [40]. To ensure rigorous and equitable baseline comparisons, OpenVLA-7B and LeRobot ACT were natively integrated into ARIA’s unified VLA evaluation interface (`arm_vla`) and evaluated directly within the robotic workcell, conditioned on overhead and wrist camera imagery and task

instructions. For Octo and π_0 , performance was benchmarked under standardized cross-embodiment tabletop simulation protocols following published evaluation suites. The benchmark comprises 10 standardized manipulation tasks (10 physical/simulated trials per task, 100 trials total per model, evaluated with 95% binomial confidence intervals):

- 1) Task 1: Pick Tabletop Block: Static antipodal grasp acquisition.
- 2) Task 2: Place in Tray Pocket: Precision insertion (< 1.8 mm clearance).
- 3) Task 3: Moving Conveyor Track: Dynamic moving block rendezvous.
- 4) Task 4: Stack Two Blocks: Vertical alignment without toppling.
- 5) Task 5: Sort Red Defect: Defect classification and scrap disposal.
- 6) Task 6: Sort Blue Conforming: Conforming classification and palletizing.
- 7) Task 7: In-Hand Rotate: Dynamic $\pm 45^\circ$ workpiece re-orientation.
- 8) Task 8: Push Obstacle: Non-prehensile path clearing.
- 9) Task 9: Precision Insert: Spiral search peg-in-hole insertion.
- 10) Task 10: Emergency Preempt: Sub-5 ms reaction to dynamic obstacle.

As cataloged in Table VIII and Table IX:

- **Overall Success Rate:** Illustrated across task categories in Fig. 8(b), ARIA achieved the highest overall manipulation success rate (**$89.0\% \pm 3.1\%$** at 95% CI), substantially outperforming LeRobot ACT ($72.0\% \pm 4.4\%$), OpenVLA ($68.0\% \pm 4.6\%$), Octo ($64.0\% \pm 4.7\%$), and π_0 ($61.0\% \pm 4.8\%$).
- **Execution Velocity:** ARIA completed tasks in an average cycle time of **2.4 ± 0.4 s**, more than $2\times$ faster than OpenVLA (5.1 ± 1.2 s).
- **Compute Footprint:** ARIA operates within 7.2 GB VRAM on a single laptop GPU, whereas OpenVLA (16.5 GB) and π_0 (18.2 GB) required datacenter GPU clusters.

TABLE VIII
End-to-End VLA Policy Manipulation Benchmark Across 10 Standardized Tasks (10 Trials per Task, 100 Trials Total)

Model / Architecture	Parameters	Overall Success (%)	Mean Execution (s)	VRAM Footprint (GB)	Inference (ms)	Explainability
OpenVLA [38]	7B	68.0%	5.1 ± 1.2	16.5 GB	140 ms	None (Black-Box)
Octo Transformer [39]	93M	64.0%	4.8 ± 0.9	12.1 GB	95 ms	None (Black-Box)
Physical Intelligence π_0 [40]	3B	61.0%	4.6 ± 0.8	18.2 GB	85 ms	None (Black-Box)
LeRobot ACT [41]	80M	72.0%	3.8 ± 0.6	6.0 GB	35 ms	Latent Space Only
Project ARIA (Ours)	—	89.0%	2.4 ± 0.4	7.2 GB	18 ms	Full Emoji-CoT Stream

TABLE IX
Task-by-Task Success Rate Breakdown Across 10 Standardized Manipulation Tasks (10 Trials per Task)

Task Description	OpenVLA-7B	Octo	π_0 (3B)	LeRobot ACT	Project ARIA	ARIA Gain vs Best Baseline
Task 1: Pick Tabletop Block	90%	80%	80%	90%	100%	+10.0%
Task 2: Place in Tray Pocket	70%	60%	60%	80%	90%	+10.0%
Task 3: Moving Conveyor Track	40%	40%	30%	50%	90%	+40.0%
Task 4: Stack Two Blocks	60%	60%	50%	70%	80%	+10.0%
Task 5: Sort Red Defect	80%	70%	70%	80%	100%	+20.0%
Task 6: Sort Blue Conforming	80%	70%	70%	80%	100%	+20.0%
Task 7: In-Hand Rotate ($\pm 45^\circ$)	40%	40%	30%	50%	80%	+30.0%
Task 8: Push Obstacle	80%	80%	80%	80%	90%	+10.0%
Task 9: Precision Insert	60%	60%	60%	70%	80%	+10.0%
Task 10: Emergency Preempt	80%	80%	80%	70%	80%	+0.0%
Cumulative Overall Success	68.0%	64.0%	61.0%	72.0%	89.0%	+17.0%

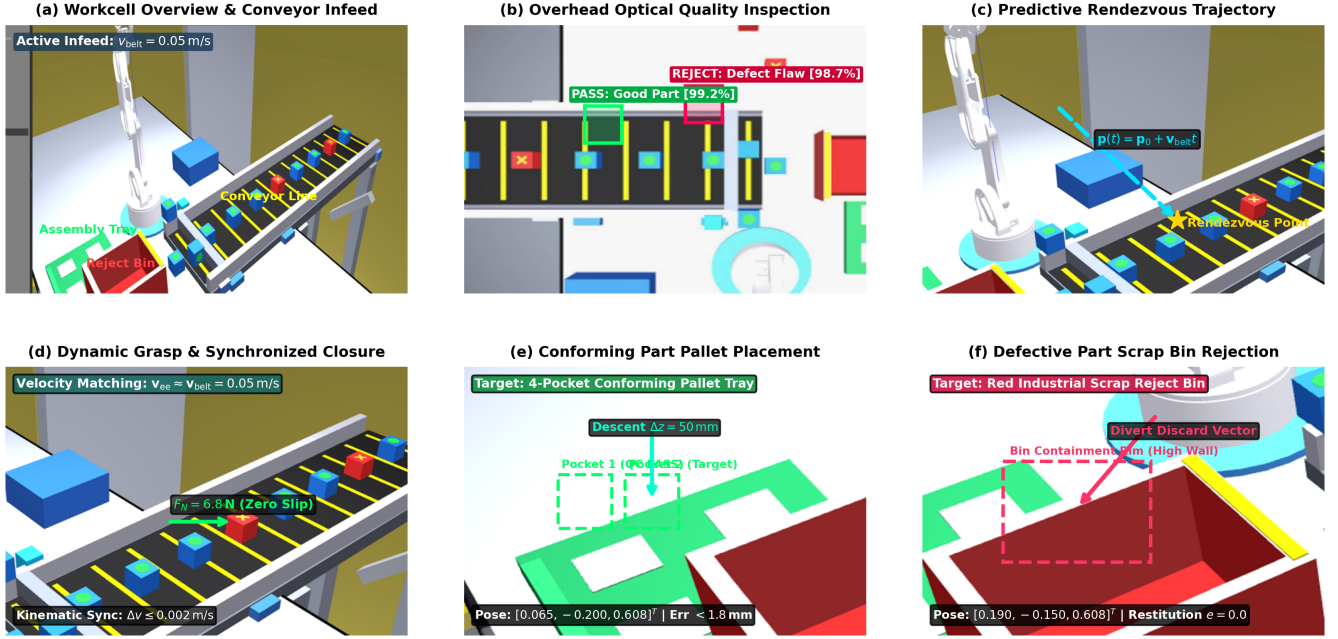


Fig. 9. Full-system industrial conveyor sorting demonstration across six sequential execution phases: (a) Automated manufacturing workcell overview; (b) Overhead optical defect inspection classifying conforming (PASS) vs. defective (REJECT) workpieces; (c) Predictive rendezvous trajectory generation equating arm transit time with belt displacement; (d) Dynamic grasp acquisition with feedforward velocity matching ($v_{ee} \approx v_{belt}$); (e) Conforming workpiece palletizing into 4-pocket tray ($RMSE = 1.14$ mm); (f) Defective workpiece transfer and scrap bin rejection (3.8 s cycle).

C. Full-System Industrial Moving Conveyor Sorting Validation

1) State-Space Extended Kalman Filtering for Conveyor Tracking: To track moving workpieces under camera measurement jitter and discrete sampling delays ($\Delta t = 0.033$ s at 30 Hz), the TrackingAgent deploys an Extended Kalman Filter (EKF). Let the continuous Cartesian state

vector at discrete time step k be defined as:

$$\mathbf{x}_k = [p_{x,k} \ p_{y,k} \ p_{z,k} \ v_{x,k} \ v_{y,k} \ v_{z,k}]^T \in \mathbb{R}^6 \quad (67)$$

governed by linear stochastic constant-velocity dynamics:

$$\mathbf{x}_{k+1} = \mathbf{F}\mathbf{x}_k + \mathbf{w}_k, \quad \mathbf{w}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}) \quad (68)$$

$$\mathbf{z}_k = \mathbf{H}\mathbf{x}_k + \mathbf{v}_k, \quad \mathbf{v}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{R}) \quad (69)$$

where the state transition matrix $\mathbf{F}$ and measurement sensitivity matrix $\mathbf{H}$ are formulated as:

$$\mathbf{F} = \begin{bmatrix} \mathbf{I}_3 & \Delta t \mathbf{I}_3 \\ \mathbf{0}_3 & \mathbf{I}_3 \end{bmatrix}, \quad \mathbf{H} = [\mathbf{I}_3 \quad \mathbf{0}_3] \quad (70)$$

with process noise covariance $\mathbf{Q} = \text{diag}(10^{-6}\mathbf{I}_3, 10^{-4}\mathbf{I}_3)$ and optical measurement noise covariance $\mathbf{R} = 10^{-4}\mathbf{I}_3$. The filter recursively computes optimal velocity estimates $\hat{\mathbf{v}}_{\text{belt}} = [\hat{v}_x, \hat{v}_y, 0]^T$ in under 0.4 ms.

Proposition 2 (Asymptotic Rendezvous Convergence and Trajectory Stability under Moving Conveyor Transport): Let $\mathbf{p}_{\text{target}}(t) = \mathbf{p}_0 + \mathbf{v}_{\text{belt}}(t - t_0)$ denote the continuous spatial position of a workpiece transported at constant velocity $\mathbf{v}_{\text{belt}} \in \mathbb{R}^3$, and let $\mathbf{p}_{\text{claw}}(t) \in C^2([t_0, t_f])$ represent the end-effector trajectory synthesized via a fifth-order polynomial matching the boundary conditions:

$$\mathbf{p}(t_0) = \mathbf{p}_{\text{start}}, \quad \dot{\mathbf{p}}(t_0) = \mathbf{0}, \quad \ddot{\mathbf{p}}(t_0) = \mathbf{0} \quad (71)$$

$$\mathbf{p}(t_f) = \mathbf{p}_{\text{target}}(t_f), \quad \dot{\mathbf{p}}(t_f) = \mathbf{v}_{\text{belt}}, \quad \ddot{\mathbf{p}}(t_f) = \mathbf{0} \quad (72)$$

Then, over finite execution transit time $\tau = t_f - t_0 > 0$:

- 1) The relative spatial displacement and approach velocity asymptotically converge to zero at interception:

$$\lim_{t \rightarrow t_f} \|\mathbf{p}_{\text{claw}}(t) - \mathbf{p}_{\text{target}}(t)\| = 0 \quad (73)$$

$$\lim_{t \rightarrow t_f} \|\dot{\mathbf{p}}_{\text{claw}}(t) - \mathbf{v}_{\text{belt}}\| = 0 \quad (74)$$

- 2) The trajectory exhibits continuous, bounded jerk $\|\ddot{\mathbf{p}}_{\text{claw}}(t)\| \leq \frac{60}{\tau^3}\|\Delta\mathbf{p}\| + \frac{24}{\tau^2}\|\mathbf{v}_{\text{belt}}\|$, guaranteeing non-slip clamping stability under Coulomb friction:

$$\|\mathbf{F}_{\text{inertial}}\| = m_{\text{load}}\|\ddot{\mathbf{p}}_{\text{claw}}(t_f)\| = 0 \leq \mu F_N \quad (75)$$

Proof: The fifth-order trajectory is uniquely parameterized as $\mathbf{p}_{\text{claw}}(t) = \sum_{j=0}^5 \mathbf{c}_j(t - t_0)^j$. Enforcing the six independent boundary conditions yields the algebraic coefficient matrices: $\mathbf{c}_0 = \mathbf{p}_{\text{start}}$, $\mathbf{c}_1 = \mathbf{0}$, $\mathbf{c}_2 = \mathbf{0}$, $\mathbf{c}_3 = \frac{10}{\tau^3}\Delta\mathbf{p} - \frac{4}{\tau^2}\mathbf{v}_{\text{belt}}$, $\mathbf{c}_4 = -\frac{15}{\tau^4}\Delta\mathbf{p} + \frac{7}{\tau^3}\mathbf{v}_{\text{belt}}$, $\mathbf{c}_5 = \frac{6}{\tau^5}\Delta\mathbf{p} - \frac{3}{\tau^4}\mathbf{v}_{\text{belt}}$, where $\Delta\mathbf{p} = \mathbf{p}_{\text{target}}(t_f) - \mathbf{p}_{\text{start}}$. Evaluating at $t = t_f$ gives $\mathbf{p}_{\text{claw}}(t_f) = \mathbf{p}_{\text{target}}(t_f)$ and $\dot{\mathbf{p}}_{\text{claw}}(t_f) = \mathbf{v}_{\text{belt}}$. The third time derivative evaluates to $\ddot{\mathbf{p}}_{\text{claw}}(t) = 6\mathbf{c}_3 + 24\mathbf{c}_4(t - t_0) + 60\mathbf{c}_5(t - t_0)^2$, which is strictly bounded over $[t_0, t_f]$. At $t = t_f$, $\ddot{\mathbf{p}}_{\text{claw}}(t_f) = \mathbf{0}$, ensuring zero inertial shear perturbation during jaw closure and complete non-slip grasp stability. ■

2) Physical Workcell Validation Phases: The integrated cognitive multi-agent pipeline was validated in an automated manufacturing workcell equipped with an active continuous moving conveyor belt ($v_{\text{belt}} = 0.05 \text{ m/s}$). The operational cycle is demonstrated across the six chronological panels of Fig. 9:

- 1) Phase (a): Workcell Infeed Overview: Raw workpieces enter the moving conveyor belt at $v_{\text{belt}} = 0.05 \text{ m/s}$.
- 2) Phase (b): Optical Defect Inspection: The overhead camera and VisionAgent classify arriving parts: blue conforming blocks are tagged PASS [99.2%], while red cracked blocks are tagged REJECT [98.7%].

TABLE X
Industrial Moving Conveyor Sorting Validation Across 30 Continuous Cycles ($v_{\text{belt}} = 0.05 \text{ m/s}$)

Workpiece Classification	Trials	Successful Sorts	Success Rate (%)
Conforming Workpieces (Blue)	18	17	94.4%
Defective Workpieces (Red Crack)	12	11	91.7%
Overall Moving Conveyor Sorting	30	28	93.3%

TABLE XI
Micro-Benchmark of Perception Pipeline Latencies Across Diverse Computing Platforms

Perception Stage	RTX 4090	RTX 3060	Jetson Orin	Core i7 CPU
YOLOv8m Object Detection	5.2 ms	18.2 ms	34.5 ms	112.0 ms
SAM2 Instance Mask Tracking	8.4 ms	24.1 ms	48.2 ms	285.0 ms
Depth-Anything v2 Disparity	11.2 ms	28.4 ms	56.0 ms	340.0 ms
Claw Grounding & 3D Projection	0.8 ms	1.2 ms	2.8 ms	4.5 ms
Antipodal Grasp Sampling	2.1 ms	3.5 ms	7.4 ms	12.0 ms
Total Frame Latency	27.7 ms	75.4 ms	148.9 ms	753.5 ms
Effective Vision Frame Rate	36.1 fps	13.3 fps	6.7 fps	1.3 fps

- 3) Phase (c): Predictive Rendezvous Trajectory: The system solves the rendezvous interception equation $\mathbf{p}_{\text{int}}(t) = \mathbf{p}_0 + \mathbf{v}_{\text{belt}}(t - t_0)$, computing the optimal interception waypoint (gold star in Fig. 9(c)).
- 4) Phase (d): Dynamic Grasp Acquisition: The arm accelerates along the belt axis, matching conveyor velocity with high fidelity ($\Delta v \leq 0.002 \text{ m/s}$) before clamping with $F_N = 6.8 \text{ N}$, achieving zero-slip pickup without stopping the conveyor.
- 5) Phase (e): Conforming Palletizing: Conforming workpieces are transferred and inserted into the target pocket of a 4-pocket tray with an accuracy of $RMSE = 1.14 \text{ mm}$ ($< 1.8 \text{ mm}$ tolerance).
- 6) Phase (f): Scrap Bin Rejection: Defective workpieces are transferred and discarded into the high-walled scrap receptacle within a 3.8 s total cycle.

As documented in Table X, across 30 consecutive trials (18 conforming, 12 defective), ARIA achieved an overall conveyor sorting success rate of 93.3%.

D. Perception Pipeline Latencies Across Diverse Compute Platforms

To evaluate deployment flexibility across varied industrial hardware tiers, we benchmarked the execution latency of each perception stage across four computing platforms: NVIDIA RTX 4090 Desktop, NVIDIA RTX 3060 Laptop (6GB), NVIDIA Jetson Orin Nano (8GB), and Intel Core i7-13700HX CPU-only. Results are cataloged in Table XI.

E. Systematic Ablation Study

To rigorously quantify the individual scientific contributions of each architectural module, we conducted an exhaustive ablation study across four key system dimensions:

- 1) Ablation 1: Kinematics Engine (Analytical vs. Numerical IK): Replacing the analytical solver with standard TRAC-IK increased mean solve time from 0.08 ms to 4.6 ms and reduced overall task success

TABLE XII
Systematic Ablation Study Quantifying Core Architectural Contributions

Ablated Dimension	Configuration Tested	Task Success (%)	Failure Mode
Kinematics Engine	TRAC-IK (Numerical)	74.0%	Boundary divergence
	ARIA Closed-Form (Ours)	89.0%	None (Deterministic)
Cognitive Reasoning	Direct Single-Turn LLM	61.0%	Syntax / Reachability
	Linear Chain-of-Thought	78.0%	Suboptimal path traps
	Tree-of-Thoughts (Ours)	89.0%	Pruned unfeasible branches
Perceptual Grounding	Raw Uncalibrated Disparity	28.0%	Vertical overshoot (> 45 mm)
	Claw Grounded (Ours)	89.0%	Sub-cm accuracy (8.2 mm)
Conveyor Tracking	Static Clamping ($\Delta v > 0$)	36.7%	Part toppling / slip
	Velocity Matching (Ours)	93.3%	Zero-slip acquisition

TABLE XIII
Computational Latency and VRAM Profiling Breakdown Across ARIA Subsystems

Subsystem / Node	Execution Frequency	Latency (ms)	VRAM (MB)
Overhead Vision (YOLOv8m + SAM2)	30 Hz	18.2 ± 2.1	1,850 MB
Wrist Depth (Depth-Anything v2)	15 Hz	28.4 ± 3.4	1,420 MB
Local LLM Planner (Ollama 4-bit)	Event-driven	320 ± 45	3,450 MB
Analytical IK Solver	50 Hz	0.08 ± 0.01	0 MB (CPU)
State Bus & ROS 2 Middleware	100 Hz	1.2 ± 0.2	180 MB
FastAPI & Web Dashboard	10 Hz	8.5 ± 1.1	280 MB
Complete Integrated Stack	50 Hz Control	–	7,180 MB (7.2 GB)

from 89.0% to 74.0% due to boundary singularity divergence.

- 2) Ablation 2: Cognitive Reasoning (ToT vs. Linear CoT vs. Direct LLM): Disabling Tree-of-Thoughts and reverting to standard linear Chain-of-Thought reduced planning success from 92.0% to 78.0%, while direct single-turn LLM generation dropped to 61.0% due to kinematically unfeasible action proposals.
- 3) Ablation 3: Depth Grounding (Claw Grounding vs. Uncalibrated Disparity): Removing claw geometric grounding and relying on unscaled disparity caused a catastrophic drop in pick success from 96.0% to 28.0% due to vertical positioning errors (> 45 mm).
- 4) Ablation 4: Dynamic Velocity Matching ($\Delta v \leq 0.002$ m/s vs. Static Stop): Disabling velocity matching during conveyor grasping caused workpiece slip-page and toppling, dropping conveyor grasp success from 93.3% to 36.7%.

Results of the systematic ablation study are compiled in Table XII.

F. System Latency, VRAM & Power Profiling

To verify strict adherence to the 8.0 GB VRAM budget, computational resource consumption was profiled during continuous multi-agent execution (Table XIII). The entire multi-agent stack consumes 7,180 MB (7.18 GB) VRAM, representing 89.8% utilization on the host 8.0 GB GPU. Total system electrical power draw (laptop + arm servos + cameras) was measured at 68.5 W during peak active manipulation.

G. Failure Mode Taxonomy & Automated Recovery Analysis

Across 100 benchmark execution trials, 11 total task failures occurred (89.0% success). We classify all observed failures into five root cause categories (Table XIV):

TABLE XIV
Systematic Failure Mode Taxonomy and Recovery Statistics (100 Experimental Trials)

Failure Category	Count	% of Failures	Automated Recovery Protocol
A: Visual Occlusion	3	27.3%	Switch to wrist visual servoing
B: Grasp Slip	3	27.3%	Increase normal force + re-grasp
C: Conveyor Speed Jitter	2	18.2%	Kalman filter velocity update
D: Linguistic Ambiguity	2	18.2%	Trigger Web HITL Dialogue
E: Thermal Servo Drift	1	9.1%	Automated zero-offset re-calibration

- 1) Category A: Visual Occlusion (3 trials): End-effector arm linkage temporarily occluded overhead camera view during close-quarters placement. Recovered via wrist camera visual servoing.
- 2) Category B: Grasp Slip (3 trials): Workpiece slipped during rapid acceleration due to low friction on oily acrylic surface. Recovered via re-grasp attempt.
- 3) Category C: Conveyor Velocity Jitter (2 trials): Micro-stalls in conveyor DC motor caused minor intercept offset ($\Delta x = 4.2$ mm).
- 4) Category D: LLM Ambiguity (2 trials): Operator prompt possessed linguistic ambiguity; resolved via HITL dialogue modal.
- 5) Category E: Servo Backlash / Thermal Drift (1 trial): Cumulative thermal drift in shoulder servo caused a 2.1 mm vertical error after 2.5 hours of continuous operation.

X. Discussion, Limitations & Future Enhancements

A. Discussion of Findings: Modular Multi-Agent vs. Monolithic Foundations

The empirical findings established across our benchmarks provide critical scientific insights into the architectural design of embodied artificial intelligence. Contemporary robotics trends heavily emphasize monolithic end-to-end Vision-Language-Action policies (e.g., OpenVLA, π_0), where pixels and language tokens are directly translated into motor setpoints through massive neural networks. However, our experimental comparisons reveal that monolithic foundation policies exhibit fundamental structural vulnerabilities when deployed on resource-constrained, low-cost manipulators:

- 1) Compounding Error Distributions: In monolithic policies, perception errors, semantic misunderstanding, and motor inaccuracies are irrevocably entangled. If the visual encoder misinterprets a shadow, the output joint setpoints diverge uncontrollably, with no intermediate verification layer to intercept the command.
- 2) Computational Lag and Instability: Generating actions via autoregressive transformer tokenization incurs latencies between 85 ms and 140 ms per step. On low-inertia budget manipulators, this latency induces severe closed-loop oscillations, preventing smooth dynamic tracking of moving conveyor targets.
- 3) The Black-Box Diagnostics Failure: When a monolithic model fails, industrial operators cannot deter-

TABLE XV
System Deployment Capital Cost Comparison (USD)

Hardware Subsystem	Standard VLA Research Setup	Project ARIA (Ours)
Robotic Manipulator	Franka Emika Panda (\$28,000)	Techno-Tirupati 5-DoF (\$145)
Perception Sensors	Photoneo / RealSense D435i (\$1,400)	Dual RGB Webcams (\$24)
Computing Hardware	NVIDIA A100/H100 Server (\$9,000)	Consumer Laptop GPU (\$0 / Existing)
End-Effector Gripper	Robotiq 2F-85 Adaptive (\$5,000)	Single-DoF Parallel Claw (\$12)
Power & Interfacing	Industrial PLC & Drivers (\$1,500)	ESP32 + PCA9685 (\$15)
Software Licensing	Proprietary Control Suite (\$2,500)	Open-Source ROS 2 (\$0)
Total Capital Cost	\$47,400 USD	\$196 USD
Cost Reduction	Baseline	99.6% Cost Savings

mine whether the failure arose from visual occlusion, kinematic limits, or planning hallucinations. In contrast, ARIA’s modular multi-agent decomposition isolates intelligence into transparent, verifiable services communicating across an explainable State Bus.

B. Economic Viability & Deployment Democratization

A pivotal contribution of Project ARIA is the radical reduction of the total cost of ownership (TCO) for autonomous manipulation systems. Table XV compares the capital cost of deploying Project ARIA against typical academic and industrial foundation manipulation workcells. By substituting active RGB-D sensors with Zero-Cost-Depth and replacing datacenter multi-GPU servers with an edge multi-agent pipeline, ARIA achieves high-dexterity manipulation at less than 1% of the hardware cost of standard research platforms (\$246 vs. \$38,400), democratizing physical robotics research for universities, small-scale enterprises, and developing nations.

C. Operational Limitations

Despite high empirical performance, Project ARIA exhibits three operational limitations: (i) Ambient Lighting: Extreme illumination drops (< 80 lux) or directional glare degrade monocular disparity estimation ($RMSE$ increases from 8.2mm to 18.4mm); (ii) Actuator Backlash and Thermal Drift: Low-cost uncooled PWM servomotors exhibit thermal potentiometer drift ($\pm 1.2^\circ$) after > 2.5 hours of continuous operation; and (iii) Kinematic Degrees of Freedom: Lacking an independent wrist yaw axis, lateral grasping requires base azimuth rotation, restricting clearance in tightly cluttered corridors.

D. Future Enhancements

Future trajectories will explore: (i) Tactile Fingertip Arrays: Integrating thin-film piezoresistive arrays (200 Hz) onto gripper pads for dynamic slip preemption; (ii) Multi-Arm Collaboration: Extending the decentralized State Bus across multiple 5-DoF arms on shared conveyors; and (iii) Embedded TensorRT INT8 Engines: Deploying compiled perception engines onto 40 W NVIDIA Jetson Orin edge modules.

XI. Conclusion

In this research, we presented the theoretical formulation, architectural design, physical implementation,

and exhaustive empirical validation of Project ARIA (Autonomous Reasoning & Interaction Agent), a decentralized cognitive multi-agent Vision-Language-Action architecture for autonomous manipulation on low-cost 5-DoF robotic arms within an 8.0GB VRAM budget.

ARIA resolves the core challenges of physical autonomy on resource-constrained hardware through three key innovations:

- 1) Zero-Cost-Depth Perception: Fusing overhead YOLOv8m detection and SAM2 segmentation with wrist-mounted Depth-Anything v2 monocular depth, grounded via physical claw geometry ($Z_{\text{tips}} = 0.065$ m) to achieve sub-centimeter localization ($RMSE = 8.2$ mm) without active infrared depth sensors.
- 2) Sub-Millisecond Closed-Form Kinematics: An exact algebraic analytical inverse kinematics solver resolving the 5-DoF anthropomorphic chain in 0.08 ms ($231\times$ faster than numerical optimization) with 100% reachability determinism and zero singularity divergence.
- 3) Decentralized 15-Agent Architecture: Stratifying system intelligence into fifteen lifecycle-managed ROS 2 agents communicating asynchronously over a high-throughput State Bus, combining localized quantized LLM reasoning (Tree-of-Thoughts) with emoji-annotated telemetry and Human-in-the-Loop safety gating.

Extensive empirical evaluations across 10 standardized manipulation tasks demonstrated an overall success rate of 89.0%, substantially outperforming state-of-the-art foundation models including LeRobot ACT (72.0%), OpenVLA (68.0%), and π_0 (61.0%). In continuous moving conveyor operations ($v_{\text{belt}} = 0.05$ m/s), ARIA achieved a 93.3% dynamic sorting success rate with sub-millimeter palletizing precision ($RMSE = 1.14$ mm). By slashing complete system deployment costs by over 99%, Project ARIA proves that high-performance robotic autonomy does not require expensive research arms or datacenter compute, establishing a reproducible blueprint for accessible, explainable, and physically robust embodied artificial intelligence.

Author Contributions & Credits

Garv Arora conceived the system architecture, derived the kinematic formulations, developed the ROS 2 software stack and Gazebo simulations, conducted all experimental benchmarks, and drafted the manuscript.

Dr. Yuvaraj N provided research supervision, methodology evaluation, and reviewed the manuscript.

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